In [2]:
```
#This code mostly follows ModelBuildExample.ipynb, written by Professor Schuckers w

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import pylab as py
import numpy as np
from sklearn import tree
from sklearn.svm import SVC
from sklearn.model_selection import cross_val_score
# here are some of the tools we will use for our analyses
from sklearn.linear_model import LinearRegression
from sklearn.metrics import root_mean_squared_error
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, precision_score, reca
from sklearn.tree import DecisionTreeClassifier
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis, QuadraticDisc
from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import PredictionErrorDisplay
```

In [3]:
```
hlt = pd.read_csv('NPA_HLT.csv', encoding='utf-8-sig')

hlt
```

Out[3]:

	OBJECTID	NPAs_NPA	NPAs_LandArea	NPA_Lookup__NPA	NPA_Lookup__Access_Scor
0	24922	188	365	188.0	2.67357
1	24923	131	270	131.0	2.80314
2	24924	64	381	64.0	3.26384
3	24925	326	152	326.0	6.98000
4	24926	336	1522	336.0	3.55777
...	...	...	...	...	.
2302	27224	304	67	304.0	5.83000
2303	27225	100	725	100.0	3.11176
2304	27226	172	944	172.0	3.02004
2305	27227	328	735	328.0	5.19418
2306	27228	344	262	344.0	6.84638

2307 rows × 26 columns

In [4]:
```
newhlt = hlt[["NPAs_NPA", "NPA_Lookup__Median_Sales_Price", "NPA_Lookup__Jobs_Acces
newhlt
```

Out[4]:

	NPAs_NPA	NPA_Lookup__Median_Sales_Price	NPA_Lookup__Jobs_Accessible_by_	NPA_
0	188	354750	973881.0	
1	131	215000	1032030.0	
2	64	189500	1176104.0	
3	326	522500	1248511.0	
4	336	240000	1230696.0	
...	...	...	...	
2302	304	0	1162519.0	
2303	100	174000	1111120.0	
2304	172	419500	1069479.0	
2305	328	152500	1241316.0	
2306	344	280000	1127778.0	

2307 rows × 4 columns

In [5]:
```
newhlt = newhlt.groupby('NPAs_NPA')[['NPA_Lookup__Median_Sales_Price','NPA_Lookup__
newhlt
```

Out[5]:

	NPAs_NPA	NPA_Lookup__Median_Sales_Price	NPA_Lookup__Jobs_Accessible_by_	NPA_L
0	2	424750.0	1179306.8	
1	3	542600.0	1226974.0	
2	4	1817800.0	1055251.0	
3	5	192300.0	1271680.2	
4	6	361200.0	1271447.6	
...	...	...	...	
458	472	273000.0	259901.0	
459	473	117500.0	553363.0	
460	474	125000.0	426426.0	
461	475	172500.0	524313.0	
462	476	390000.0	1252043.0	

463 rows × 4 columns

In [6]:
```
newhlt.columns = ['NPA', 'median_sale_price', 'jobs_accessable_by_45_minutes', 'med
```

In [7]:
```
crm = pd.read_csv('CrimeViolent.csv', encoding='utf-8-sig')

crm
```

Out[7]:

	NPA	2011	2013	2015	2016	2017	2018	2019	2020	2021	...	2011 Raw	2013 Raw	201 Ra
0	2	9	7	5	5	5	7	8	9	8	...	19	15	1
1	3	2	2	2	4	4	4	3	4	3	...	13	17	1
2	4	0	0	1	0	0	0	0	0	1	...	0	0	
3	5	22	25	19	20	16	12	26	25	19	...	13	19	1
4	6	27	31	24	29	38	28	30	32	28	...	45	55	4
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
454	472	0	0	0	1	--	3	--	--	--	...	0	0	
455	473	--	2	4	3	1	1	2	0	3	...	0	5	
456	474	2	1	1	5	--	3	--	--	--	...	4	3	
457	475	--	0	0	0	0	0	0	0	1	...	0	0	
458	476	13	13	12	13	13	9	10	12	11	...	157	186	19

459 rows × 21 columns

In [8]:
```
newcrm = crm[["NPA", "2020 Raw"]]
```

In [9]:
```
newcrm.columns = ['NPA', 'crime_rate']
newcrm
```

Out[9]:

	NPA	crime_rate
0	2	19
1	3	43
2	4	0
3	5	19
4	6	57
...	...	...
454	472	0
455	473	0
456	474	0
457	475	0
458	476	198

459 rows × 2 columns

In [10]:
```
emp = pd.read_csv('Employment.csv', encoding='utf-8-sig')
emp.columns = ['NPA', 'employment_rate']
emp
```

Out[10]:

	NPA	employment_rate
0	2	95.6%
1	3	97.4%
2	4	94.3%
3	5	82.8%
4	6	100%
...	...	...
454	472	100%
455	473	99.8%
456	474	100%
457	475	94%
458	476	96.9%

459 rows × 2 columns

In [11]:
```
pop = pd.read_csv('PopulationDensity.csv', encoding='utf-8-sig')
```

```
pop
```

Out[11]:

	NPA	2000	2010	2013	2015	2016	2017	2018	2020	2000 Raw	2010 Raw	2013 Raw	201 Ra
0	2	6	5	5	5	5	5	5	5	2,413	2,039	2,167	2,21
1	3	5	6	7	7	7	8	8	10	5,694	6,563	7,810	7,93
2	4	5	3	3	3	3	3	3	4	1,535	1,002	1,065	1,07
3	5	4	4	5	4	5	5	5	5	659	603	763	74
4	6	5	4	4	4	5	5	5	4	2,024	1,638	1,775	1,72
...	...	...	...	...	...	...	...	...	...	...	...	...	
454	472	2	3	3	3	4	4	4	4	900	1,140	1,290	1,31
455	473	3	4	5	5	5	5	5	5	1,340	1,879	2,235	2,27
456	474	4	4	5	5	5	5	5	5	1,949	2,173	2,382	2,39
457	475	2	2	3	3	3	3	3	3	1,513	1,775	2,626	2,71
458	476	4	9	10	11	11	12	13	12	5,237	12,489	14,251	15,43

459 rows × 17 columns

In [12]:

```
newpop = pop[["NPA", "2020"]]
newpop.columns = ['NPA', 'population_density']
newpop
```

Out[12]:

	NPA	population_density
0	2	5
1	3	10
2	4	4
3	5	5
4	6	4
...	...	...
454	472	4
455	473	5
456	474	5
457	475	3
458	476	12

459 rows × 2 columns

In [13]:
```
df = pd.merge(newhlt, newcrm, on='NPA')
df
```

Out[13]:

	NPA	median_sale_price	jobs_accessable_by_45_minutes	median_household_income	cri
0	2	424750.0	1179306.8	51095.0	
1	3	542600.0	1226974.0	101143.2	
2	4	1817800.0	1055251.0	186159.2	
3	5	192300.0	1271680.2	29648.0	
4	6	361200.0	1271447.6	30749.8	
...	...	...	...	...	
454	472	273000.0	259901.0	89063.0	
455	473	117500.0	553363.0	50953.4	
456	474	125000.0	426426.0	49949.8	
457	475	172500.0	524313.0	117029.4	
458	476	390000.0	1252043.0	93764.0	

459 rows × 5 columns

In [14]:
```
df = pd.merge(df, emp, on='NPA')
df
```

Out[14]:

	NPA	median_sale_price	jobs_accessable_by_45_minutes	median_household_income	cri
0	2	424750.0	1179306.8	51095.0	
1	3	542600.0	1226974.0	101143.2	
2	4	1817800.0	1055251.0	186159.2	
3	5	192300.0	1271680.2	29648.0	
4	6	361200.0	1271447.6	30749.8	
...	...	...	...	...	
454	472	273000.0	259901.0	89063.0	
455	473	117500.0	553363.0	50953.4	
456	474	125000.0	426426.0	49949.8	
457	475	172500.0	524313.0	117029.4	
458	476	390000.0	1252043.0	93764.0	

459 rows × 6 columns

In [15]:
```
df = pd.merge(df, newpop, on='NPA')
df
```

Out[15]:

	NPA	median_sale_price	jobs_accessable_by_45_minutes	median_household_income	cri
0	2	424750.0	1179306.8	51095.0	
1	3	542600.0	1226974.0	101143.2	
2	4	1817800.0	1055251.0	186159.2	
3	5	192300.0	1271680.2	29648.0	
4	6	361200.0	1271447.6	30749.8	
...	...	...	...	...	
454	472	273000.0	259901.0	89063.0	
455	473	117500.0	553363.0	50953.4	
456	474	125000.0	426426.0	49949.8	
457	475	172500.0	524313.0	117029.4	
458	476	390000.0	1252043.0	93764.0	

459 rows × 7 columns

In [16]:
```
df['employment_rate'] = pd.to_numeric(df['employment_rate'].str.replace('%', ''), e
```

In [17]:
```
df.head()
```

Out[17]:

	NPA	median_sale_price	jobs_accessable_by_45_minutes	median_household_income	crim
0	2	424750.0	1179306.8	51095.0	
1	3	542600.0	1226974.0	101143.2	
2	4	1817800.0	1055251.0	186159.2	
3	5	192300.0	1271680.2	29648.0	
4	6	361200.0	1271447.6	30749.8	

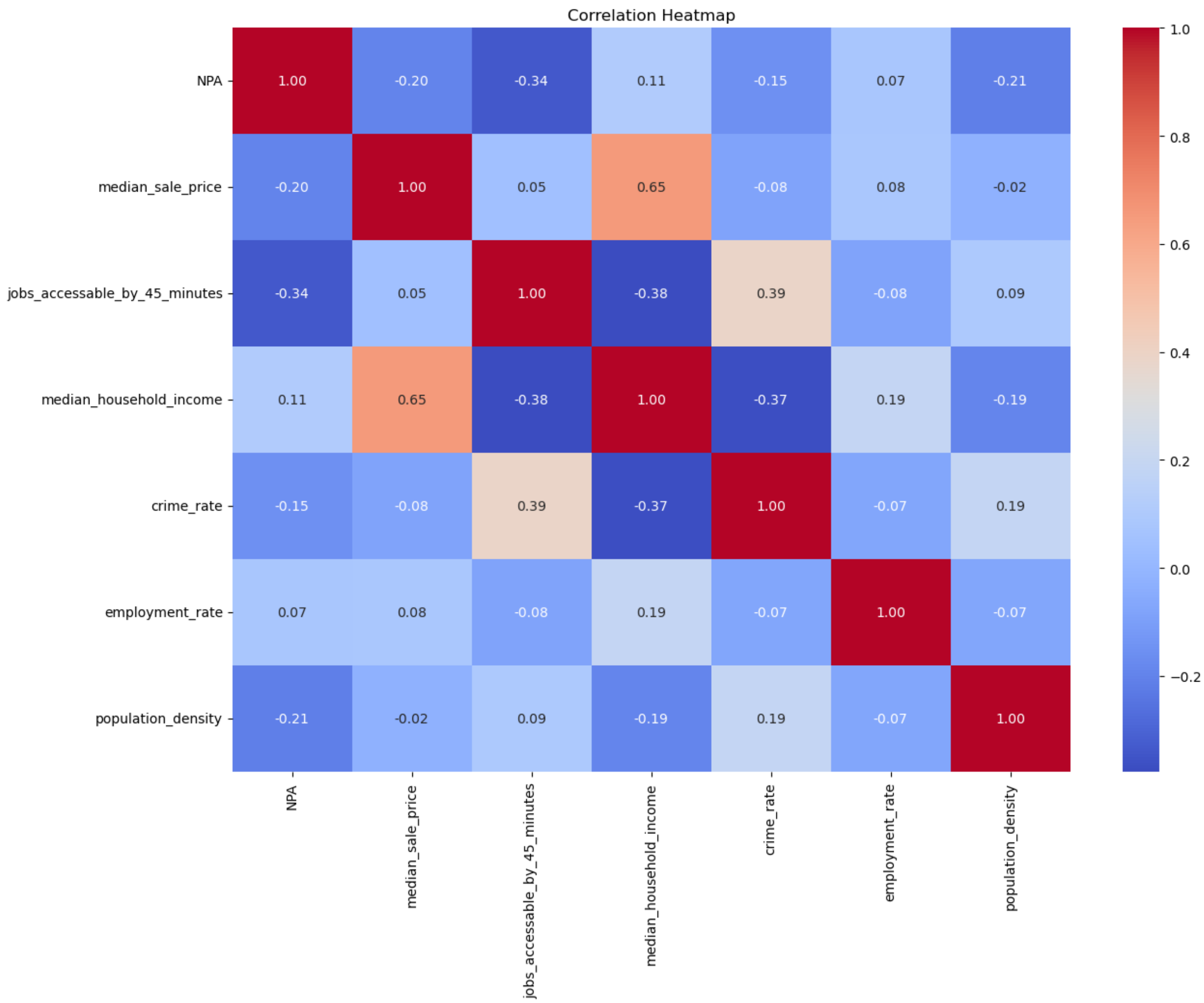
In [18]:
```
print(f"\nTotal rows with any NaN: {df.isnull().any(axis=1).sum()}")
print(f"Total rows: {len(df)}")
```

```
Total rows with any NaN: 2
Total rows: 459
```

In [19]:
```
df = df.dropna()
```

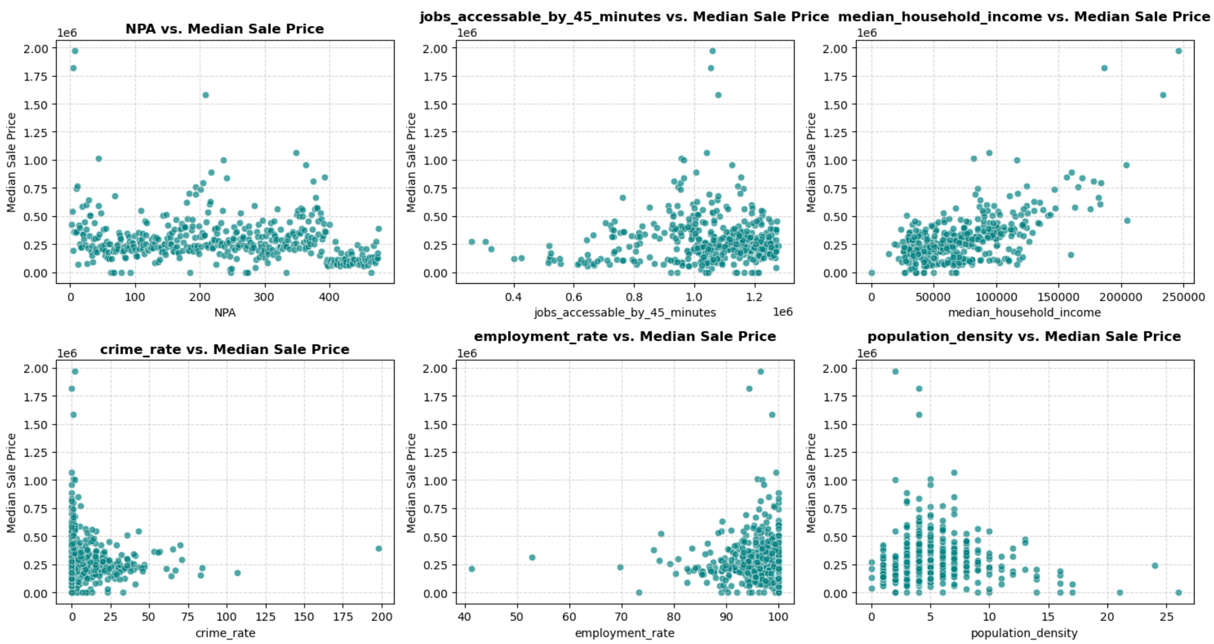
In [20]:
```
for col in df.columns:
    count = (df[col].astype(str) == '--').sum()
    if count > 0:
        print(f"{col}: {count} occurrences of '--'")
```

In [21]:
```
corr = df.corr(numeric_only=True)

# plot and show the correlation in a heatmap

plt.figure(figsize=(14, 10))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```



In [22]:
```
cols_to_plot = df.select_dtypes(include=['number']).columns.drop('median_sale_price

# We define the layout (how many plots per row)
plots_per_row = 3
# Calculate necessary rows based on the number of features
num_rows = (len(cols_to_plot) + plots_per_row - 1) // plots_per_row

# Create the figure and subplots
fig, axes = plt.subplots(nrows=num_rows, ncols=plots_per_row, figsize=(15, 4 * num_
```

```
# Flatten axes array for easy iteration, regardless of layout
axes_flat = axes.flatten()

# Loop through each column and generate its scatterplot
for i, col in enumerate(cols_to_plot):
    # sns.scatterplot handles the plotting on the specified axis (ax=axes_flat[i])
    sns.scatterplot(data=df, x=col, y='median_sale_price', ax=axes_flat[i], color='

    # Customizing each plot
    axes_flat[i].set_title(f'{col} vs. Median Sale Price', fontsize=12, fontweight=
    axes_flat[i].set_xlabel(col, fontsize=10)
    axes_flat[i].set_ylabel('Median Sale Price', fontsize=10)
    axes_flat[i].grid(True, linestyle='--', alpha=0.5)

# Hide any empty subplots if the number of features isn't a perfect multiple
for i in range(len(cols_to_plot), len(axes_flat)):
    fig.delaxes(axes_flat[i])

# Optimize the layout spacing
plt.tight_layout()

# Display all plots at once
plt.show()
```



In [23]:
```
df['log_crime_rate']=np.log(df['crime_rate']+1)
df['log_employment_rate']=np.log(df['employment_rate']+1)
df['log_jobs_accessable_by_45_minutes']=np.log(df['jobs_accessable_by_45_minutes']+
df.columns
```

Out[23]:
```
Index(['NPA', 'median_sale_price', 'jobs_accessable_by_45_minutes',
       'median_household_income', 'crime_rate', 'employment_rate',
       'population_density', 'log_crime_rate', 'log_employment_rate',
       'log_jobs_accessable_by_45_minutes'],
      dtype='str')
```

In [24]:
```
cols_to_plot = df.select_dtypes(include=['number']).columns.drop('median_sale_price
```

```
# We define the layout (how many plots per row)
plots_per_row = 3
# Calculate necessary rows based on the number of features
num_rows = (len(cols_to_plot) + plots_per_row - 1) // plots_per_row

# Create the figure and subplots
fig, axes = plt.subplots(nrows=num_rows, ncols=plots_per_row, figsize=(15, 4 * num_

# Flatten axes array for easy iteration, regardless of layout
axes_flat = axes.flatten()

# Loop through each column and generate its scatterplot
for i, col in enumerate(cols_to_plot):
    # sns.scatterplot handles the plotting on the specified axis (ax=axes_flat[i])
    sns.scatterplot(data=df, x=col, y='median_sale_price', ax=axes_flat[i], color='

    # Customizing each plot
    axes_flat[i].set_title(f'{col} vs. Median Sale Price', fontsize=12, fontweight=
    axes_flat[i].set_xlabel(col, fontsize=10)
    axes_flat[i].set_ylabel('Median Sale Price', fontsize=10)
    axes_flat[i].grid(True, linestyle='--', alpha=0.5)

# Hide any empty subplots if the number of features isn't a perfect multiple
for i in range(len(cols_to_plot), len(axes_flat)):
    fig.delaxes(axes_flat[i])

# Optimize the layout spacing
plt.tight_layout()

# Display all plots at once
plt.show()
```

In [25]:
```
# selecting columns and target variable
X=df[['jobs_accessable_by_45_minutes', 'median_household_income', 'crime_rate', 'em
y = df['median_sale_price']

# splitting the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

In [26]:
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [27]:
```
# Create a linear regression model
full_model = LinearRegression()

# Fit the model on the  data
full_model.fit(X, y)

# Make predictions on the  data
y_hat = full_model.predict(X)

# Evaluate the model
rmse = root_mean_squared_error(y, y_hat)
print('Root Mean Squared Error:', rmse)

x2 = sm.add_constant(X)

#fit linear regression model
```

```
full_model_a = sm.OLS(y, x2).fit()

#view model summary
print(full_model_a.summary())
```

```
Root Mean Squared Error: 143060.3667885444
                            OLS Regression Results
==============================================================================
Dep. Variable:      median_sale_price   R-squared:                       0.539
Model:                            OLS   Adj. R-squared:                  0.534
Method:                 Least Squares   F-statistic:                     105.6
Date:                Mon, 04 May 2026   Prob (F-statistic):           1.26e-73
Time:                        03:38:51   Log-Likelihood:                -6073.5
No. Observations:                 457   AIC:                         1.216e+04
Df Residuals:                     451   BIC:                         1.218e+04
Df Model:                           5
Covariance Type:            nonrobust
=====================================================================================
=============
                                   coef    std err          t      P>|t|      [0.02
5      0.975]
-------------------------------------------------------------------------------------
-------------
const                        -3.381e+05   1.27e+05     -2.662      0.008   -5.88e+0
5   -8.85e+04
jobs_accessable_by_45_minutes     0.3642      0.042      8.756      0.000       0.28
2       0.446
median_household_income           4.8324      0.214     22.530      0.000       4.41
1       5.254
crime_rate                      981.0018    449.272      2.184      0.030      98.07
6    1863.928
employment_rate               -1432.3154   1235.560     -1.159      0.247   -3860.48
4     995.853
population_density             5466.0411   2079.682      2.628      0.009    1378.97
1    9553.111
==============================================================================
Omnibus:                      214.270   Durbin-Watson:                   1.682
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2025.188
Skew:                           1.783   Prob(JB):                         0.00
Kurtosis:                      12.676   Cond. No.                     2.02e+07
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
[2] The condition number is large, 2.02e+07. This might indicate that there are
strong multicollinearity or other numerical problems.
```

In [ ]:

In [28]:

```
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error
# Create a regression tree model
dtree = DecisionTreeRegressor(random_state=0)

# Train the model
```

```
dtree.fit(X_train, y_train)

# Make predictions
predictions = dtree.predict(X_test)

# Evaluate the model
rmse = np.sqrt(mean_squared_error(y_test, predictions))
print(f"Mean Squared Error: {rmse}")
```

```
Mean Squared Error: 168018.25995810048
```

In [29]:
```
# create a plot of the decision tree
fig = plt.figure(figsize=(25,20))
tree.plot_tree(dtree,
                   feature_names=['jobs_accessable_by_45_minutes', 'median_househol
                   filled=True)
```

```
Out[29]: [Text(0.7666685072931448, 0.9821428571428571, 'median_household_income <= 160051.8
98\nsquared_error = 43637489026.196\nsamples = 365\nvalue = 286998.904'),
 Text(0.546573905525569, 0.9464285714285714, 'median_household_income <= 82806.797
\nsquared_error = 24331450451.391\nsamples = 356\nvalue = 267830.758'),
 Text(0.6566212064093568, 0.9642857142857142, 'True  '),
 Text(0.27433924613924016, 0.9107142857142857, 'crime_rate <= 0.5\nsquared_error =
10786629705.547\nsamples = 258\nvalue = 218214.729'),
 Text(0.06262244411377751, 0.875, 'jobs_accessable_by_45_minutes <= 1027640.5\nsqu
ared_error = 3796305844.444\nsamples = 45\nvalue = 108626.667'),
 Text(0.025554553341108475, 0.8392857142857143, 'jobs_accessable_by_45_minutes <=
553895.5\nsquared_error = 830474225.0\nsamples = 20\nvalue = 82345.0'),
 Text(0.008824593959519473, 0.8035714285714286, 'median_household_income <= 5859
4.398\nsquared_error = 343055555.556\nsamples = 3\nvalue = 108333.333'),
 Text(0.005883062639679649, 0.7678571428571429, 'employment_rate <= 99.9\nsquared_
error = 14062500.0\nsamples = 2\nvalue = 121250.0'),
 Text(0.0029415313198398246, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1
\nvalue = 117500.0'),
 Text(0.008824593959519473, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1
\nvalue = 125000.0'),
 Text(0.011766125279359298, 0.7678571428571429, 'squared_error = 0.0\nsamples = 1
\nvalue = 82500.0'),
 Text(0.042284512722697475, 0.8035714285714286, 'median_household_income <= 7371
6.598\nsquared_error = 776269186.851\nsamples = 17\nvalue = 77758.824'),
 Text(0.025738399048598464, 0.7678571428571429, 'median_household_income <= 4343
7.0\nsquared_error = 716558668.639\nsamples = 13\nvalue = 71430.769'),
 Text(0.014707656599199122, 0.7321428571428571, 'population_density <= 3.5\nsquare
d_error = 310570000.0\nsamples = 4\nvalue = 86700.0'),
 Text(0.011766125279359298, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1
\nvalue = 57800.0'),
 Text(0.017649187919038946, 0.6964285714285714, 'jobs_accessable_by_45_minutes <=
933034.0\nsquared_error = 42888888.889\nsamples = 3\nvalue = 96333.333'),
 Text(0.014707656599199122, 0.6607142857142857, 'employment_rate <= 97.2\nsquared_
error = 12250000.0\nsamples = 2\nvalue = 100500.0'),
 Text(0.011766125279359298, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 9700
0.0'),
 Text(0.017649187919038946, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 1040
00.0'),
 Text(0.02059071923887877, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 88000.0'),
 Text(0.03676914149799781, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 8
64541.5\nsquared_error = 747321913.58\nsamples = 9\nvalue = 64644.444'),
 Text(0.029415313198398244, 0.6964285714285714, 'population_density <= 4.0\nsquare
d_error = 179657600.0\nsamples = 5\nvalue = 81630.0'),
 Text(0.02647378187855842, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 107250.0'),
 Text(0.032356844518238066, 0.6607142857142857, 'population_density <= 6.5\nsquare
d_error = 19451875.0\nsamples = 4\nvalue = 75225.0'),
 Text(0.02647378187855842, 0.625, 'jobs_accessable_by_45_minutes <= 608819.5\nsqua
red_error = 250000.0\nsamples = 2\nvalue = 79500.0'),
 Text(0.023532250558718597, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1
\nvalue = 80000.0'),
 Text(0.029415313198398244, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1
\nvalue = 79000.0'),
 Text(0.03823990715791772, 0.625, 'employment_rate <= 98.25\nsquared_error = 21025
00.0\nsamples = 2\nvalue = 70950.0'),
 Text(0.03529837583807789, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
```

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value = 69500.0'),
 Text(0.04118143847775754, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 72400.0'),
 Text(0.04412296979759737, 0.6964285714285714, 'jobs_accessable_by_45_minutes <= 9
40285.0\nsquared_error = 645470468.75\nsamples = 4\nvalue = 43412.5'),
 Text(0.04118143847775754, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 0.0'),
 Text(0.047064501117437194, 0.6607142857142857, 'median_household_income <= 5467
4.4\nsquared_error = 23007222.222\nsamples = 3\nvalue = 57883.333'),
 Text(0.04412296979759737, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 5110
0.0'),
 Text(0.05000603243727701, 0.625, 'median_household_income <= 67484.0\nsquared_err
or = 625.0\nsamples = 2\nvalue = 61275.0'),
 Text(0.047064501117437194, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1
\nvalue = 61250.0'),
 Text(0.05294756375711684, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 61300.0'),
 Text(0.05883062639679649, 0.7678571428571429, 'population_density <= 2.5\nsquared
_error = 417216875.0\nsamples = 4\nvalue = 98325.0'),
 Text(0.05294756375711684, 0.7321428571428571, 'median_household_income <= 77702.0
\nsquared_error = 33062500.0\nsamples = 2\nvalue = 116250.0'),
 Text(0.05000603243727701, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 122000.0'),
 Text(0.05588909507695666, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 110500.0'),
 Text(0.06471368903647613, 0.7321428571428571, 'employment_rate <= 93.9\nsquared_e
rror = 158760000.0\nsamples = 2\nvalue = 80400.0'),
 Text(0.061772157716636314, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1
\nvalue = 67800.0'),
 Text(0.06765522035631596, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 93000.0'),
 Text(0.09969033488644655, 0.8392857142857143, 'population_density <= 13.0\nsquare
d_error = 5174325696.0\nsamples = 25\nvalue = 129652.0'),
 Text(0.09674880356660673, 0.8035714285714286, 'jobs_accessable_by_45_minutes <= 1
160887.812\nsquared_error = 4660337482.639\nsamples = 24\nvalue = 135054.167'),
 Text(0.0938072722467669, 0.7678571428571429, 'median_household_income <= 39231.20
1\nsquared_error = 3155540812.854\nsamples = 23\nvalue = 126619.565'),
 Text(0.07647981431583543, 0.7321428571428571, 'employment_rate <= 96.2\nsquared_e
rror = 1298302222.222\nsamples = 3\nvalue = 201066.667'),
 Text(0.07353828299599562, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 250200.0'),
 Text(0.07942134563567525, 0.6964285714285714, 'median_household_income <= 31387.8
01\nsquared_error = 136890000.0\nsamples = 2\nvalue = 176500.0'),
 Text(0.07647981431583543, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 188200.0'),
 Text(0.08236287695551509, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 164800.0'),
 Text(0.11113473017769837, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.100563601997024, 0.6964285714285714, 'median_household_income <= 75709.699
\nsquared_error = 1792854480.969\nsamples = 17\nvalue = 104441.176'),
 Text(0.08824593959519474, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 1
034300.5\nsquared_error = 367699948.98\nsamples = 14\nvalue = 92357.143'),
 Text(0.0853044082753549, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 13720
0.0'),
 Text(0.09118747091503455, 0.625, 'jobs_accessable_by_45_minutes <= 1046624.5\nsqu
```

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ared_error = 229402633.136\nsamples = 13\nvalue = 88907.692'),
 Text(0.0753767400708955, 0.5892857142857143, 'median_household_income <= 53123.4
\nsquared_error = 77775555.556\nsamples = 3\nvalue = 73433.333'),
 Text(0.07243520875105568, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 61300.0'),
 Text(0.07831827139073533, 0.5535714285714286, 'employment_rate <= 97.3\nsquared_e
rror = 6250000.0\nsamples = 2\nvalue = 79500.0'),
 Text(0.0753767400708955, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 77000.0'),
 Text(0.08125980271057515, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 82000.0'),
 Text(0.10699820175917361, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.09743822496969419, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 1
075008.5\nsquared_error = 112720585.938\nsamples = 8\nvalue = 98168.75'),
 Text(0.0871428653502548, 0.5178571428571429, 'median_household_income <= 59437.1
\nsquared_error = 37562400.0\nsamples = 5\nvalue = 92160.0'),
 Text(0.08125980271057515, 0.48214285714285715, 'population_density <= 1.5\nsquare
d_error = 7022500.0\nsamples = 2\nvalue = 87350.0'),
 Text(0.07831827139073533, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 90000.0'),
 Text(0.08420133403041498, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 84700.0'),
 Text(0.09302592798993445, 0.48214285714285715, 'median_household_income <= 7151
7.199\nsquared_error = 32215555.556\nsamples = 3\nvalue = 95366.667'),
 Text(0.09008439667009463, 0.44642857142857145, 'jobs_accessable_by_45_minutes <=
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 Text(0.0871428653502548, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 102900.0'),
 Text(0.09302592798993445, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 94000.0'),
 Text(0.09596745930977427, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 89200.0'),
 Text(0.10773358458913357, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 1
083435.875\nsquared_error = 77517222.222\nsamples = 3\nvalue = 108183.333'),
 Text(0.10479205326929375, 0.48214285714285715, 'employment_rate <= 98.4\nsquared_
error = 4950625.0\nsamples = 2\nvalue = 114275.0'),
 Text(0.10185052194945392, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 116500.0'),
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\nvalue = 112050.0'),
 Text(0.1106751159089734, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 96000.0'),
 Text(0.11655817854865304, 0.5535714285714286, 'median_household_income <= 47311.6
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value = 80550.0'),
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01\nsquared_error = 4582055555.556\nsamples = 3\nvalue = 160833.333'),
 Text(0.10993973307901343, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 25650
0.0'),
 Text(0.11582279571869308, 0.625, 'median_household_income <= 79211.102\nsquared_e
rror = 9000000.0\nsamples = 2\nvalue = 113000.0'),
 Text(0.11288126439885326, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
```

```
value = 110000.0'),
 Text(0.11876432703853292, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 116000.0'),
 Text(0.12170585835837273, 0.6964285714285714, 'population_density <= 2.5\nsquared
_error = 1780415000.0\nsamples = 3\nvalue = 177850.0'),
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_error = 642622500.0\nsamples = 2\nvalue = 203850.0'),
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0.0'),
 Text(0.09969033488644655, 0.7678571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 329050.0'),
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value = 0.0'),
 Text(0.48605604816470277, 0.875, 'median_household_income <= 57090.4\nsquared_err
or = 9190197253.301\nsamples = 213\nvalue = 241367.136'),
 Text(0.3482891397457184, 0.8392857142857143, 'population_density <= 13.5\nsquared
_error = 8698047060.046\nsamples = 137\nvalue = 221143.796'),
 Text(0.2517923161112037, 0.8035714285714286, 'jobs_accessable_by_45_minutes <= 10
64764.062\nsquared_error = 8144606557.007\nsamples = 128\nvalue = 227927.344'),
 Text(0.17538880494544953, 0.7678571428571429, 'median_household_income <= 36586.0
\nsquared_error = 8464233337.925\nsamples = 33\nvalue = 184278.788'),
 Text(0.14817964023693114, 0.7321428571428571, 'median_household_income <= 34368.5
\nsquared_error = 8980811358.025\nsamples = 9\nvalue = 106444.444'),
 Text(0.14523810891709132, 0.6964285714285714, 'population_density <= 7.0\nsquared
_error = 7384562448.98\nsamples = 7\nvalue = 136857.143'),
 Text(0.13641351495757187, 0.6607142857142857, 'crime_rate <= 4.5\nsquared_error =
1248948888.889\nsamples = 3\nvalue = 223266.667'),
 Text(0.13347198363773202, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 17340
0.0'),
 Text(0.1393550462774117, 0.625, 'employment_rate <= 97.45\nsquared_error = 841000
0.0\nsamples = 2\nvalue = 248200.0'),
 Text(0.13641351495757187, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 251100.0'),
 Text(0.1422965775972515, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 245300.0'),
 Text(0.1540627028766108, 0.6607142857142857, 'median_household_income <= 32592.9
\nsquared_error = 2186352500.0\nsamples = 4\nvalue = 72050.0'),
 Text(0.151121171556771, 0.625, 'employment_rate <= 86.15\nsquared_error = 7714466
66.667\nsamples = 3\nvalue = 48900.0'),
 Text(0.14817964023693114, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 88000.0'),
 Text(0.1540627028766108, 0.5892857142857143, 'crime_rate <= 17.5\nsquared_error =
10562500.0\nsamples = 2\nvalue = 29350.0'),
 Text(0.151121171556771, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nva
lue = 26100.0'),
 Text(0.15700423419645063, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 32600.0'),
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0.0'),
 Text(0.151121171556771, 0.6964285714285714, 'squared_error = 0.0\nsamples = 2\nva
lue = 0.0'),
 Text(0.2025979696539679, 0.7321428571428571, 'population_density <= 7.5\nsquared_
```

```
error = 5146764930.556\nsamples = 24\nvalue = 213466.667'),
 Text(0.18200725041508914, 0.6964285714285714, 'population_density <= 1.5\nsquared
_error = 2408205955.556\nsamples = 15\nvalue = 180923.333'),
 Text(0.17906571909524932, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 69000.0'),
 Text(0.18494878173492896, 0.6607142857142857, 'median_household_income <= 45793.3
98\nsquared_error = 1621534502.551\nsamples = 14\nvalue = 188917.857'),
 Text(0.17171189079564975, 0.625, 'employment_rate <= 89.75\nsquared_error = 20105
78877.551\nsamples = 7\nvalue = 165257.143'),
 Text(0.1658288281559701, 0.5892857142857143, 'crime_rate <= 13.5\nsquared_error =
25871666.667\nsamples = 3\nvalue = 212200.0'),
 Text(0.1628872968361303, 0.5535714285714286, 'crime_rate <= 8.5\nsquared_error =
1000000.0\nsamples = 2\nvalue = 215750.0'),
 Text(0.15994576551629044, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 216750.0'),
 Text(0.1658288281559701, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 214750.0'),
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value = 205100.0'),
 Text(0.1775949534353294, 0.5892857142857143, 'crime_rate <= 7.5\nsquared_error =
606842500.0\nsamples = 4\nvalue = 130050.0'),
 Text(0.17465342211548956, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 92200.0'),
 Text(0.18053648475516923, 0.5535714285714286, 'crime_rate <= 15.0\nsquared_error
= 172402222.222\nsamples = 3\nvalue = 142666.667'),
 Text(0.1775949534353294, 0.5178571428571429, 'population_density <= 6.5\nsquared_
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 Text(0.17465342211548956, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1
\nvalue = 127900.0'),
 Text(0.18053648475516923, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1
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 Text(0.18347801607500905, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 159800.0'),
 Text(0.19818567267420817, 0.625, 'jobs_accessable_by_45_minutes <= 1014382.281\ns
quared_error = 112831326.531\nsamples = 7\nvalue = 212578.571'),
 Text(0.18936107871468869, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 9
86577.625\nsquared_error = 18975555.556\nsamples = 3\nvalue = 201933.333'),
 Text(0.18641954739484887, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 195800.0'),
 Text(0.19230261003452853, 0.5535714285714286, 'crime_rate <= 10.5\nsquared_error
= 250000.0\nsamples = 2\nvalue = 205000.0'),
 Text(0.18936107871468869, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 204500.0'),
 Text(0.19524414135436835, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 205500.0'),
 Text(0.20701026663372765, 0.5892857142857143, 'median_household_income <= 55071.0
\nsquared_error = 34489218.75\nsamples = 4\nvalue = 220562.5'),
 Text(0.20406873531388783, 0.5535714285714286, 'employment_rate <= 91.95\nsquared_
error = 535555.556\nsamples = 3\nvalue = 223933.333'),
 Text(0.201127203994048, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nva
lue = 222900.0'),
 Text(0.20701026663372765, 0.5178571428571429, 'population_density <= 5.5\nsquared
_error = 2500.0\nsamples = 2\nvalue = 224450.0'),
 Text(0.20406873531388783, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1
\nvalue = 224500.0'),
 Text(0.20995179795356747, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1
```

```
\nvalue = 224400.0'),
 Text(0.20995179795356747, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 210450.0'),
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 Text(0.21730562625316702, 0.6607142857142857, 'employment_rate <= 98.05\nsquared_
error = 699602500.0\nsamples = 2\nvalue = 351550.0'),
 Text(0.2143640949333272, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 32510
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 Text(0.22024715757300686, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 37800
0.0'),
 Text(0.22907175153252632, 0.6607142857142857, 'median_household_income <= 51019.8
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 Text(0.2261302202126865, 0.625, 'median_household_income <= 37954.898\nsquared_er
ror = 760811180.556\nsamples = 6\nvalue = 221391.667'),
 Text(0.21877639191308695, 0.5892857142857143, 'median_household_income <= 37855.1
\nsquared_error = 255200625.0\nsamples = 2\nvalue = 190675.0'),
 Text(0.2158348605932471, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
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 Text(0.22171792323292677, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 206650.0'),
 Text(0.23348404851228607, 0.5892857142857143, 'crime_rate <= 13.5\nsquared_error
= 305981250.0\nsamples = 4\nvalue = 236750.0'),
 Text(0.2276009858726064, 0.5535714285714286, 'population_density <= 9.0\nsquared_
error = 3330625.0\nsamples = 2\nvalue = 224125.0'),
 Text(0.2246594545527666, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 225950.0'),
 Text(0.23054251719244623, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 222300.0'),
 Text(0.2393671111519657, 0.5535714285714286, 'crime_rate <= 23.0\nsquared_error =
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 Text(0.2364255798321259, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 266400.0'),
 Text(0.24230864247180553, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 232350.0'),
 Text(0.23201328285236616, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 37790
0.0'),
 Text(0.3281958272769578, 0.7678571428571429, 'population_density <= 0.5\nsquared_
error = 7141883047.091\nsamples = 95\nvalue = 243089.474'),
 Text(0.32231276463727815, 0.7321428571428571, 'crime_rate <= 2.5\nsquared_error =
2256250000.0\nsamples = 2\nvalue = 85100.0'),
 Text(0.31937123331743833, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 132600.0'),
 Text(0.32525429595711797, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\n
value = 37600.0'),
 Text(0.3340788899166375, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 12
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 Text(0.33113735859679766, 0.6964285714285714, 'employment_rate <= 93.0\nsquared_e
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 Text(0.2627155160031943, 0.6607142857142857, 'employment_rate <= 79.8\nsquared_er
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 Text(0.24230864247180553, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 209950.0'),
 Text(0.2481917051114852, 0.5892857142857143, 'employment_rate <= 65.0\nsquared_er
```

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ror = 694267222.222\nsamples = 3\nvalue = 282183.333'),
 Text(0.24525017379164535, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\n
value = 315400.0'),
 Text(0.251133236431325, 0.5535714285714286, 'crime_rate <= 28.0\nsquared_error =
213890625.0\nsamples = 2\nvalue = 265575.0'),
 Text(0.2481917051114852, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 280200.0'),
 Text(0.25407476775116483, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 250950.0'),
 Text(0.28018085821474326, 0.625, 'jobs_accessable_by_45_minutes <= 1071844.562\ns
quared_error = 2054492422.521\nsamples = 22\nvalue = 210520.455'),
 Text(0.27723932689490344, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 289600.0'),
 Text(0.2831223895345831, 0.5892857142857143, 'median_household_income <= 30301.10
1\nsquared_error = 1840355691.61\nsamples = 21\nvalue = 206754.762'),
 Text(0.2628993617106843, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 12
40254.5\nsquared_error = 4321194400.0\nsamples = 5\nvalue = 232190.0'),
 Text(0.25995783039084447, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 355400.0'),
 Text(0.2658408930305241, 0.5178571428571429, 'employment_rate <= 86.1\nsquared_er
ror = 657522968.75\nsamples = 4\nvalue = 201387.5'),
 Text(0.25995783039084447, 0.48214285714285715, 'median_household_income <= 2874
3.0\nsquared_error = 204490000.0\nsamples = 2\nvalue = 178000.0'),
 Text(0.25701629907100465, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 163700.0'),
 Text(0.2628993617106843, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 192300.0'),
 Text(0.2717239556702038, 0.48214285714285715, 'employment_rate <= 89.95\nsquared_
error = 16605625.0\nsamples = 2\nvalue = 224775.0'),
 Text(0.268782424350364, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\nv
alue = 228850.0'),
 Text(0.2746654869900436, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 220700.0'),
 Text(0.3033454173584819, 0.5535714285714286, 'median_household_income <= 34770.4
\nsquared_error = 799742460.938\nsamples = 16\nvalue = 198806.25'),
 Text(0.2893731435892427, 0.5178571428571429, 'median_household_income <= 32298.2
\nsquared_error = 759258750.0\nsamples = 4\nvalue = 163500.0'),
 Text(0.2834900809495631, 0.48214285714285715, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.28054854962972325, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 195750.0'),
 Text(0.2864316122694029, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 182600.0'),
 Text(0.2952562062289224, 0.48214285714285715, 'median_household_income <= 33995.4
\nsquared_error = 156875625.0\nsamples = 2\nvalue = 137825.0'),
 Text(0.29231467490908253, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 150350.0'),
 Text(0.2981977375487622, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 125300.0'),
 Text(0.31731769112772107, 0.5178571428571429, 'population_density <= 3.5\nsquared
_error = 259223125.0\nsamples = 12\nvalue = 210575.0'),
 Text(0.3070223315082817, 0.48214285714285715, 'employment_rate <= 83.15\nsquared_
error = 53024400.0\nsamples = 5\nvalue = 197040.0'),
 Text(0.30408080018844186, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 209950.0'),
 Text(0.3099638628281215, 0.44642857142857145, 'crime_rate <= 33.5\nsquared_error
```

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= 14196718.75\nsamples = 4\nvalue = 193812.5'),
 Text(0.3070223315082817, 0.4107142857142857, 'jobs_accessable_by_45_minutes <= 11
91154.125\nsquared_error = 367222.222\nsamples = 3\nvalue = 195966.667'),
 Text(0.30408080018844186, 0.375, 'crime_rate <= 15.0\nsquared_error = 50625.0\nsa
mples = 2\nvalue = 196375.0'),
 Text(0.30113926886860204, 0.3392857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 196150.0'),
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alue = 196600.0'),
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0.0'),
 Text(0.3129053941479613, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 187350.0'),
 Text(0.32761305074716046, 0.48214285714285715, 'population_density <= 8.0\nsquare
d_error = 182186020.408\nsamples = 7\nvalue = 220242.857'),
 Text(0.3217299881074808, 0.44642857142857145, 'population_density <= 4.5\nsquared
_error = 19455000.0\nsamples = 4\nvalue = 230700.0'),
 Text(0.31878845678764095, 0.4107142857142857, 'squared_error = 0.0\nsamples = 2\n
value = 234800.0'),
 Text(0.32467151942732064, 0.4107142857142857, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.3217299881074808, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 22890
0.0'),
 Text(0.32761305074716046, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 22430
0.0'),
 Text(0.3334961133868401, 0.44642857142857145, 'median_household_income <= 38511.1
99\nsquared_error = 58955000.0\nsamples = 3\nvalue = 206300.0'),
 Text(0.3305545820670003, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 216100.0'),
 Text(0.3364376447066799, 0.4107142857142857, 'population_density <= 9.5\nsquared_
error = 16402500.0\nsamples = 2\nvalue = 201400.0'),
 Text(0.3334961133868401, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 20545
0.0'),
 Text(0.33937917602651974, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 19735
0.0'),
 Text(0.39955920119040095, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 1
270698.0\nsquared_error = 7500316235.652\nsamples = 66\nvalue = 254262.879'),
 Text(0.37639392389936743, 0.625, 'employment_rate <= 93.95\nsquared_error = 71582
46179.138\nsamples = 63\nvalue = 248640.476'),
 Text(0.3584991296054786, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 11
93750.688\nsquared_error = 13494656020.408\nsamples = 7\nvalue = 322107.143'),
 Text(0.35114530130587907, 0.5535714285714286, 'median_household_income <= 45007.6
99\nsquared_error = 5777341600.0\nsamples = 5\nvalue = 261980.0'),
 Text(0.3452622386661994, 0.5178571428571429, 'crime_rate <= 55.5\nsquared_error =
1783482222.222\nsamples = 3\nvalue = 207066.667'),
 Text(0.34232070734635955, 0.48214285714285715, 'jobs_accessable_by_45_minutes <=
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 Text(0.33937917602651974, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1
\nvalue = 243700.0'),
 Text(0.3452622386661994, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 229600.0'),
 Text(0.3482037699860392, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 147900.0'),
 Text(0.3570283639455587, 0.5178571428571429, 'median_household_income <= 50992.80
1\nsquared_error = 460102500.0\nsamples = 2\nvalue = 344350.0'),
 Text(0.3540868326257189, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
```

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value = 322900.0'),
 Text(0.3599698952653985, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 365800.0'),
 Text(0.36585295790507816, 0.5535714285714286, 'population_density <= 6.5\nsquared
_error = 1154300625.0\nsamples = 2\nvalue = 472425.0'),
 Text(0.36291142658523834, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 506400.0'),
 Text(0.368794489224918, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nva
lue = 438450.0'),
 Text(0.3942887181932563, 0.5892857142857143, 'median_household_income <= 28112.6
\nsquared_error = 5607192448.98\nsamples = 56\nvalue = 239457.143'),
 Text(0.3776190831844375, 0.5535714285714286, 'employment_rate <= 95.25\nsquared_e
rror = 3549469600.0\nsamples = 5\nvalue = 164070.0'),
 Text(0.3746775518645976, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 275400.0'),
 Text(0.3805606145042773, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 12
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 Text(0.3776190831844375, 0.48214285714285715, 'median_household_income <= 27155.2
\nsquared_error = 81946666.667\nsamples = 3\nvalue = 123300.0'),
 Text(0.3746775518645976, 0.44642857142857145, 'jobs_accessable_by_45_minutes <= 1
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alue = 113100.0'),
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alue = 121700.0'),
 Text(0.3805606145042773, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 135100.0'),
 Text(0.3835021458241171, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 175050.0'),
 Text(0.41095835320207513, 0.5535714285714286, 'population_density <= 1.5\nsquared
_error = 5197126074.587\nsamples = 51\nvalue = 246848.039'),
 Text(0.4080168218822353, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 424650.0'),
 Text(0.413899884521915, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 113
1872.5\nsquared_error = 4656152436.0\nsamples = 50\nvalue = 243292.0'),
 Text(0.3937975054435565, 0.48214285714285715, 'employment_rate <= 94.9\nsquared_e
rror = 2287418163.265\nsamples = 7\nvalue = 295357.143'),
 Text(0.38644367714395694, 0.44642857142857145, 'crime_rate <= 12.5\nsquared_error
= 24502500.0\nsamples = 2\nvalue = 227000.0'),
 Text(0.3835021458241171, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 231950.0'),
 Text(0.38938520846379676, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 222050.0'),
 Text(0.40115133374315604, 0.44642857142857145, 'jobs_accessable_by_45_minutes <=
1117423.312\nsquared_error = 575873000.0\nsamples = 5\nvalue = 322700.0'),
 Text(0.3952682711034764, 0.4107142857142857, 'jobs_accessable_by_45_minutes <= 11
02054.125\nsquared_error = 44410555.556\nsamples = 3\nvalue = 304733.333'),
 Text(0.3923267397836366, 0.375, 'employment_rate <= 98.85\nsquared_error = 564062
5.0\nsamples = 2\nvalue = 300225.0'),
 Text(0.38938520846379676, 0.3392857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 297850.0'),
 Text(0.3952682711034764, 0.3392857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 302600.0'),
 Text(0.3982098024233162, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 31375
0.0'),
 Text(0.40703439638283573, 0.4107142857142857, 'jobs_accessable_by_45_minutes <= 1
```

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130719.188\nsquared_error = 162562500.0\nsamples = 2\nvalue = 349650.0'),
 Text(0.40409286506299585, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 33690
0.0'),
 Text(0.40997592770267555, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 36240
0.0'),
 Text(0.4340022636002735, 0.48214285714285715, 'jobs_accessable_by_45_minutes <= 1
144294.688\nsquared_error = 4528632874.527\nsamples = 43\nvalue = 234816.279'),
 Text(0.4217420529820348, 0.44642857142857145, 'population_density <= 8.0\nsquared
_error = 15218967222.222\nsamples = 3\nvalue = 169983.333'),
 Text(0.418800521662195, 0.4107142857142857, 'crime_rate <= 43.0\nsquared_error =
1157700625.0\nsamples = 2\nvalue = 254975.0'),
 Text(0.4158589903423552, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 22095
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 Text(0.4217420529820348, 0.375, 'squared_error = 0.0\nsamples = 1\nvalue = 28900
0.0'),
 Text(0.42468358430187464, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 0.0'),
 Text(0.4462624742185121, 0.44642857142857145, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.4433209428986723, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 412850.0'),
 Text(0.44920400553835194, 0.4107142857142857, 'population_density <= 7.5\nsquared
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 Text(0.42762511562171446, 0.375, 'jobs_accessable_by_45_minutes <= 1264546.625\ns
quared_error = 1206892920.11\nsamples = 33\nvalue = 225286.364'),
 Text(0.42468358430187464, 0.3392857142857143, 'median_household_income <= 51357.7
99\nsquared_error = 1107974606.934\nsamples = 32\nvalue = 223251.562'),
 Text(0.39623346106779883, 0.30357142857142855, 'median_household_income <= 5083
7.0\nsquared_error = 1159584328.922\nsamples = 23\nvalue = 216345.652'),
 Text(0.393291929747959, 0.26785714285714285, 'employment_rate <= 95.95\nsquared_e
rror = 987711740.702\nsamples = 22\nvalue = 219470.455'),
 Text(0.3634170022808358, 0.23214285714285715, 'employment_rate <= 94.95\nsquared_
error = 929286835.938\nsamples = 8\nvalue = 200218.75'),
 Text(0.35606317398123627, 0.19642857142857142, 'jobs_accessable_by_45_minutes <=
1239696.625\nsquared_error = 351771666.667\nsamples = 3\nvalue = 235100.0'),
 Text(0.3531216426613964, 0.16071428571428573, 'employment_rate <= 94.45\nsquared_
error = 18705625.0\nsamples = 2\nvalue = 222075.0'),
 Text(0.3501801113415566, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 22640
0.0'),
 Text(0.35606317398123627, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 21775
0.0'),
 Text(0.3590047053010761, 0.16071428571428573, 'squared_error = 0.0\nsamples = 1\n
value = 261150.0'),
 Text(0.37077083058043536, 0.19642857142857142, 'crime_rate <= 23.5\nsquared_error
= 107762400.0\nsamples = 5\nvalue = 179290.0'),
 Text(0.3648877679407557, 0.16071428571428573, 'employment_rate <= 95.3\nsquared_e
rror = 20162222.222\nsamples = 3\nvalue = 186666.667'),
 Text(0.3619462366209159, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 19300
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 Text(0.36782929926059554, 0.125, 'population_density <= 5.5\nsquared_error = 1600
00.0\nsamples = 2\nvalue = 183500.0'),
 Text(0.3648877679407557, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 183900.0'),
 Text(0.37077083058043536, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1
\nvalue = 183100.0'),
 Text(0.376653893220115, 0.16071428571428573, 'employment_rate <= 95.5\nsquared_er
```

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ror = 35105625.0\nsamples = 2\nvalue = 168225.0'),
 Text(0.3737123619002752, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 17415
0.0'),
 Text(0.3795954245399548, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 16230
0.0'),
 Text(0.42316685721508224, 0.23214285714285715, 'population_density <= 4.5\nsquare
d_error = 688288469.388\nsamples = 14\nvalue = 230471.429'),
 Text(0.410849194813253, 0.19642857142857142, 'median_household_income <= 44104.0
\nsquared_error = 162347222.222\nsamples = 12\nvalue = 220916.667'),
 Text(0.3950384639691139, 0.16071428571428573, 'crime_rate <= 13.5\nsquared_error
= 72857098.765\nsamples = 9\nvalue = 215161.111'),
 Text(0.3854784871796345, 0.125, 'crime_rate <= 10.0\nsquared_error = 1102500.0\ns
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 Text(0.3825369558597947, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1\n
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 Text(0.38842001849947433, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1
\nvalue = 223700.0'),
 Text(0.40459844075859336, 0.125, 'crime_rate <= 29.0\nsquared_error = 59582040.81
6\nsamples = 7\nvalue = 212421.429'),
 Text(0.39430308113915397, 0.08928571428571429, 'employment_rate <= 99.05\nsquared
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 Text(0.39136154981931415, 0.05357142857142857, 'squared_error = 0.0\nsamples = 1
\nvalue = 197950.0'),
 Text(0.3972446124589938, 0.05357142857142857, 'median_household_income <= 34460.2
\nsquared_error = 950625.0\nsamples = 2\nvalue = 209675.0'),
 Text(0.39430308113915397, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 208700.0'),
 Text(0.4001861437788336, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 210650.0'),
 Text(0.41489380037803275, 0.08928571428571429, 'jobs_accessable_by_45_minutes <=
1238495.812\nsquared_error = 22755468.75\nsamples = 4\nvalue = 217412.5'),
 Text(0.4090107377383531, 0.05357142857142857, 'crime_rate <= 42.5\nsquared_error
= 22500.0\nsamples = 2\nvalue = 222150.0'),
 Text(0.40606920641851324, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 222000.0'),
 Text(0.41195226905819293, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 222300.0'),
 Text(0.4207768630177124, 0.05357142857142857, 'median_household_income <= 31507.3
99\nsquared_error = 600625.0\nsamples = 2\nvalue = 212675.0'),
 Text(0.41783533169787257, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 211900.0'),
 Text(0.4237183943375522, 0.017857142857142856, 'squared_error = 0.0\nsamples = 1
\nvalue = 213450.0'),
 Text(0.426659925657392, 0.16071428571428573, 'employment_rate <= 97.85\nsquared_e
rror = 33300555.556\nsamples = 3\nvalue = 238183.333'),
 Text(0.4237183943375522, 0.125, 'median_household_income <= 47076.602\nsquared_er
ror = 4515625.0\nsamples = 2\nvalue = 242075.0'),
 Text(0.4207768630177124, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1\n
value = 244200.0'),
 Text(0.426659925657392, 0.08928571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 239950.0'),
 Text(0.42960145697723184, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 23040
0.0'),
 Text(0.4354845196169115, 0.19642857142857142, 'median_household_income <= 33984.3
99\nsquared_error = 9610000.0\nsamples = 2\nvalue = 287800.0'),
 Text(0.43254298829707166, 0.16071428571428573, 'squared_error = 0.0\nsamples = 1
```

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\nvalue = 284700.0'),
 Text(0.43842605093675135, 0.16071428571428573, 'squared_error = 0.0\nsamples = 1
\nvalue = 290900.0'),
 Text(0.39917499238763865, 0.26785714285714285, 'squared_error = 0.0\nsamples = 1
\nvalue = 147600.0'),
 Text(0.45313370753595045, 0.30357142857142855, 'crime_rate <= 6.5\nsquared_error
= 542737222.222\nsamples = 9\nvalue = 240900.0'),
 Text(0.45019217621611063, 0.26785714285714285, 'squared_error = 0.0\nsamples = 1
\nvalue = 286300.0'),
 Text(0.45607523885579027, 0.26785714285714285, 'population_density <= 3.5\nsquare
d_error = 320728750.0\nsamples = 8\nvalue = 235225.0'),
 Text(0.45019217621611063, 0.23214285714285715, 'jobs_accessable_by_45_minutes <=
1254706.375\nsquared_error = 243772500.0\nsamples = 4\nvalue = 245250.0'),
 Text(0.4472506448962708, 0.19642857142857142, 'jobs_accessable_by_45_minutes <= 1
181130.312\nsquared_error = 23428888.889\nsamples = 3\nvalue = 253933.333'),
 Text(0.444309113576431, 0.16071428571428573, 'squared_error = 0.0\nsamples = 1\nv
alue = 247700.0'),
 Text(0.45019217621611063, 0.16071428571428573, 'crime_rate <= 22.5\nsquared_error
= 6002500.0\nsamples = 2\nvalue = 257050.0'),
 Text(0.4472506448962708, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 25460
0.0'),
 Text(0.45313370753595045, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 25950
0.0'),
 Text(0.45313370753595045, 0.19642857142857142, 'squared_error = 0.0\nsamples = 1
\nvalue = 219200.0'),
 Text(0.4619583014954699, 0.23214285714285715, 'median_household_income <= 54349.6
\nsquared_error = 196683750.0\nsamples = 4\nvalue = 225200.0'),
 Text(0.4590167701756301, 0.19642857142857142, 'squared_error = 0.0\nsamples = 1\n
value = 248050.0'),
 Text(0.4648998328153098, 0.19642857142857142, 'median_household_income <= 56417.3
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 Text(0.4619583014954699, 0.16071428571428573, 'median_household_income <= 55319.0
\nsquared_error = 9922500.0\nsamples = 2\nvalue = 214150.0'),
 Text(0.4590167701756301, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 21100
0.0'),
 Text(0.4648998328153098, 0.125, 'squared_error = 0.0\nsamples = 1\nvalue = 21730
0.0'),
 Text(0.4678413641351496, 0.16071428571428573, 'squared_error = 0.0\nsamples = 1\n
value = 224450.0'),
 Text(0.4305666469415543, 0.3392857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 290400.0'),
 Text(0.4707828954549894, 0.375, 'population_density <= 8.5\nsquared_error = 72814
93958.333\nsamples = 6\nvalue = 289975.0'),
 Text(0.4678413641351496, 0.3392857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 449350.0'),
 Text(0.47372442677482923, 0.3392857142857143, 'employment_rate <= 97.85\nsquared_
error = 2641699000.0\nsamples = 5\nvalue = 258100.0'),
 Text(0.4678413641351496, 0.30357142857142855, 'median_household_income <= 29882.3
\nsquared_error = 183705000.0\nsamples = 3\nvalue = 298950.0'),
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value = 316500.0'),
 Text(0.4707828954549894, 0.26785714285714285, 'crime_rate <= 15.0\nsquared_error
= 44555625.0\nsamples = 2\nvalue = 290175.0'),
 Text(0.4678413641351496, 0.23214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 296850.0'),
 Text(0.47372442677482923, 0.23214285714285715, 'squared_error = 0.0\nsamples = 1
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\nvalue = 283500.0'),
 Text(0.47960748941450887, 0.30357142857142855, 'jobs_accessable_by_45_minutes <=
1188240.062\nsquared_error = 70980625.0\nsamples = 2\nvalue = 196825.0'),
 Text(0.47666595809466905, 0.26785714285714285, 'squared_error = 0.0\nsamples = 1
\nvalue = 188400.0'),
 Text(0.4825490207343487, 0.26785714285714285, 'squared_error = 0.0\nsamples = 1\n
value = 205250.0'),
 Text(0.42272447848143446, 0.625, 'employment_rate <= 98.35\nsquared_error = 79315
555.556\nsamples = 3\nvalue = 372333.333'),
 Text(0.41978294716159464, 0.5892857142857143, 'median_household_income <= 29483.3
\nsquared_error = 26010000.0\nsamples = 2\nvalue = 377900.0'),
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alue = 383000.0'),
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value = 372800.0'),
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alue = 361200.0'),
 Text(0.3370204212364773, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 454000.0'),
 Text(0.44478596338023313, 0.8035714285714286, 'median_household_income <= 28248.9
\nsquared_error = 6606907222.222\nsamples = 9\nvalue = 124666.667'),
 Text(0.4418444320603933, 0.7678571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 0.0'),
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d_error = 5247208125.0\nsamples = 8\nvalue = 140250.0'),
 Text(0.44478596338023313, 0.7321428571428571, 'employment_rate <= 97.05\nsquared_
error = 4398261938.776\nsamples = 7\nvalue = 126114.286'),
 Text(0.43743213508063356, 0.6964285714285714, 'jobs_accessable_by_45_minutes <= 1
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 Text(0.4315490724409539, 0.6607142857142857, 'crime_rate <= 25.5\nsquared_error =
1240800625.0\nsamples = 2\nvalue = 35225.0'),
 Text(0.4286075411211141, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 0.0'),
 Text(0.43449060376079374, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 7045
0.0'),
 Text(0.44331519772031325, 0.6607142857142857, 'employment_rate <= 95.9\nsquared_e
rror = 258537222.222\nsamples = 3\nvalue = 138616.667'),
 Text(0.44037366640047343, 0.625, 'employment_rate <= 94.9\nsquared_error = 242556
25.0\nsamples = 2\nvalue = 149625.0'),
 Text(0.43743213508063356, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
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 Text(0.44331519772031325, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 144700.0'),
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0.0'),
 Text(0.4521397916798327, 0.6964285714285714, 'employment_rate <= 98.9\nsquared_er
ror = 66422500.0\nsamples = 2\nvalue = 198250.0'),
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alue = 206400.0'),
 Text(0.4550813229996725, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 190100.0'),
 Text(0.45066902601991277, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\n
value = 239200.0'),
 Text(0.6238229565836871, 0.8392857142857143, 'population_density <= 7.5\nsquared_
error = 8011132197.022\nsamples = 76\nvalue = 277822.368'),
 Text(0.5782004952343746, 0.8035714285714286, 'employment_rate <= 99.1\nsquared_er
ror = 5557021649.03\nsamples = 63\nvalue = 261777.778'),
```

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 Text(0.5395354448778862, 0.7678571428571429, 'population_density <= 5.5\nsquared_
error = 3108215195.752\nsamples = 49\nvalue = 273329.592'),
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020019.0\nsquared_error = 2198429861.111\nsamples = 36\nvalue = 263566.667'),
 Text(0.463905916959192, 0.6964285714285714, 'median_household_income <= 66259.098
\nsquared_error = 1643899897.959\nsamples = 7\nvalue = 296271.429'),
 Text(0.46096438563935216, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\n
value = 388100.0'),
 Text(0.4668474482790318, 0.6607142857142857, 'crime_rate <= 6.5\nsquared_error =
278233055.556\nsamples = 6\nvalue = 280966.667'),
 Text(0.46096438563935216, 0.625, 'employment_rate <= 93.85\nsquared_error = 19623
125.0\nsamples = 4\nvalue = 291725.0'),
 Text(0.45802285431951234, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
value = 284400.0'),
 Text(0.463905916959192, 0.5892857142857143, 'employment_rate <= 96.35\nsquared_er
ror = 2317222.222\nsamples = 3\nvalue = 294166.667'),
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value = 296150.0'),
 Text(0.4668474482790318, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 10
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lue = 292450.0'),
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2500.0\nsamples = 2\nvalue = 259450.0'),
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alue = 249400.0'),
 Text(0.498009295698585, 0.6964285714285714, 'jobs_accessable_by_45_minutes <= 105
3002.0\nsquared_error = 2011783032.105\nsamples = 29\nvalue = 255672.414'),
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_error = 2513015000.0\nsamples = 3\nvalue = 172650.0'),
 Text(0.4786135735583911, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 10700
0.0'),
 Text(0.48449663619807076, 0.625, 'employment_rate <= 96.4\nsquared_error = 537080
625.0\nsamples = 2\nvalue = 205475.0'),
 Text(0.48155510487823094, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\n
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alue = 228650.0'),
 Text(0.514463486518939, 0.6607142857142857, 'crime_rate <= 2.5\nsquared_error = 1
066867592.456\nsamples = 26\nvalue = 265251.923'),
 Text(0.4962627614774301, 0.625, 'jobs_accessable_by_45_minutes <= 1089190.938\nsq
uared_error = 155353888.889\nsamples = 3\nvalue = 304383.333'),
 Text(0.4933212301575902, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 321200.0'),
 Text(0.4992042927972699, 0.5892857142857143, 'population_density <= 4.5\nsquared_
error = 20930625.0\nsamples = 2\nvalue = 295975.0'),
 Text(0.4962627614774301, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 300550.0'),
 Text(0.5021458241171097, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 291400.0'),
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8364.839\nsamples = 23\nvalue = 260147.826'),
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 Text(0.5080288867567894, 0.5535714285714286, 'employment_rate <= 95.1\nsquared_er
ror = 594763333.333\nsamples = 6\nvalue = 266900.0'),
 Text(0.5021458241171097, 0.5178571428571429, 'median_household_income <= 60136.0
\nsquared_error = 186340555.556\nsamples = 3\nvalue = 251466.667'),
 Text(0.4992042927972699, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 233350.0'),
 Text(0.5050873554369495, 0.48214285714285715, 'crime_rate <= 4.0\nsquared_error = 
33350625.0\nsamples = 2\nvalue = 260525.0'),
 Text(0.5021458241171097, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 254750.0'),
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value = 266300.0'),
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 Text(0.5109704180766292, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 250050.0'),
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alue = 311350.0'),
 Text(0.5256780746758283, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 330650.0'),
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 Text(0.5418564969349473, 0.5535714285714286, 'crime_rate <= 13.0\nsquared_error = 
234639897.959\nsamples = 7\nvalue = 262078.571'),
 Text(0.5345026686353478, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 11
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72803888.889\nsamples = 3\nvalue = 260016.667'),
 Text(0.5256780746758283, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
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 Text(0.5286196059956682, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
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alue = 266000.0'),
 Text(0.5403857312750274, 0.48214285714285715, 'employment_rate <= 94.25\nsquared_
error = 5522500.0\nsamples = 2\nvalue = 245050.0'),
 Text(0.5374441999551877, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 242700.0'),
 Text(0.5433272625948673, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 247400.0'),
 Text(0.5492103252345469, 0.5178571428571429, 'median_household_income <= 64449.90
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 Text(0.5462687939147071, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 284500.0'),
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 Text(0.5521518565543867, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
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 Text(0.563917981833746, 0.48214285714285715, 'crime_rate <= 10.5\nsquared_error =
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 Text(0.5580349191940663, 0.44642857142857145, 'crime_rate <= 5.5\nsquared_error =
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alue = 255750.0'),
 Text(0.5609764505139062, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 261400.0'),
 Text(0.5698010444734257, 0.44642857142857145, 'crime_rate <= 27.5\nsquared_error
= 15057968.75\nsamples = 4\nvalue = 244312.5'),
 Text(0.5668595131535858, 0.4107142857142857, 'crime_rate <= 21.0\nsquared_error =
643888.889\nsamples = 3\nvalue = 246516.667'),
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ed_error = 30625.0\nsamples = 2\nvalue = 247075.0'),
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alue = 237700.0'),
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3977.688\nsquared_error = 4632738224.852\nsamples = 13\nvalue = 300365.385'),
 Text(0.5881856152224246, 0.6964285714285714, 'median_household_income <= 74496.69
9\nsquared_error = 2200528223.141\nsamples = 11\nvalue = 281413.636'),
 Text(0.5771548727730252, 0.6607142857142857, 'median_household_income <= 60046.1
\nsquared_error = 1545585898.438\nsamples = 8\nvalue = 262006.25'),
 Text(0.5742133414531854, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 33960
0.0'),
 Text(0.5800964040928651, 0.625, 'median_household_income <= 62209.799\nsquared_er
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 Text(0.5727425757932655, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 11
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 Text(0.5698010444734257, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 11
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 Text(0.5668595131535858, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 203250.0'),
 Text(0.5727425757932655, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 221250.0'),
 Text(0.5756841071131054, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 246400.0'),
 Text(0.5874502323924646, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 10
94728.688\nsquared_error = 158735468.75\nsamples = 4\nvalue = 271387.5'),
 Text(0.581567169752785, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 940
657.625\nsquared_error = 15625.0\nsamples = 2\nvalue = 258825.0'),
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 Text(0.5786256384329451, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
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 Text(0.5845087010726248, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 258950.0'),
 Text(0.5933332950321443, 0.5535714285714286, 'employment_rate <= 94.95\nsquared_e
rror = 1822500.0\nsamples = 2\nvalue = 283950.0'),
 Text(0.5903917637123044, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
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 Text(0.6021578889916638, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 31020
0.0'),
 Text(0.6080409516313434, 0.6964285714285714, 'median_household_income <= 60501.0
\nsquared_error = 5169610000.0\nsamples = 2\nvalue = 404600.0'),
 Text(0.6050994203115035, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 332700.0'),
 Text(0.6109824829511832, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 476500.0'),
 Text(0.6168655455908628, 0.7678571428571429, 'jobs_accessable_by_45_minutes <= 97
6741.312\nsquared_error = 12026094808.673\nsamples = 14\nvalue = 221346.429'),
 Text(0.613924014271023, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nva
lue = 0.0'),
 Text(0.6198070769107027, 0.7321428571428571, 'crime_rate <= 1.5\nsquared_error =
8892484467.456\nsamples = 13\nvalue = 238373.077'),
 Text(0.6168655455908628, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 397300.0'),
 Text(0.6227486082305426, 0.6964285714285714, 'crime_rate <= 4.5\nsquared_error =
7353309774.306\nsamples = 12\nvalue = 225129.167'),
 Text(0.6168655455908628, 0.6607142857142857, 'median_household_income <= 62846.89
8\nsquared_error = 10190171250.0\nsamples = 4\nvalue = 173600.0'),
 Text(0.613924014271023, 0.625, 'jobs_accessable_by_45_minutes <= 1045240.125\nsqu
ared_error = 192690555.556\nsamples = 3\nvalue = 231466.667'),
 Text(0.6109824829511832, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 212250.0'),
 Text(0.6168655455908628, 0.5892857142857143, 'jobs_accessable_by_45_minutes <= 11
80286.938\nsquared_error = 12075625.0\nsamples = 2\nvalue = 241075.0'),
 Text(0.613924014271023, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nva
lue = 244550.0'),
 Text(0.6198070769107027, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 237600.0'),
 Text(0.6198070769107027, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 0.0'),
 Text(0.6286316708702222, 0.6607142857142857, 'crime_rate <= 6.0\nsquared_error =
3943437773.438\nsamples = 8\nvalue = 250893.75'),
 Text(0.6256901395503823, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 35240
0.0'),
 Text(0.631573202190062, 0.625, 'population_density <= 6.5\nsquared_error = 282457
8877.551\nsamples = 7\nvalue = 236392.857'),
 Text(0.6286316708702222, 0.5892857142857143, 'median_household_income <= 65813.4
```

```
\nsquared_error = 976358680.556\nsamples = 6\nvalue = 218191.667'),
 Text(0.6256901395503823, 0.5535714285714286, 'crime_rate <= 12.5\nsquared_error =
110556400.0\nsamples = 5\nvalue = 231490.0'),
 Text(0.6198070769107027, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 12
29975.5\nsquared_error = 30840555.556\nsamples = 3\nvalue = 239216.667'),
 Text(0.6168655455908628, 0.48214285714285715, 'population_density <= 3.0\nsquared
_error = 2560000.0\nsamples = 2\nvalue = 235400.0'),
 Text(0.613924014271023, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\nv
alue = 237000.0'),
 Text(0.6198070769107027, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 233800.0'),
 Text(0.6227486082305426, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 246850.0'),
 Text(0.631573202190062, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 121
7049.75\nsquared_error = 6250000.0\nsamples = 2\nvalue = 219900.0'),
 Text(0.6286316708702222, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 222400.0'),
 Text(0.6345147335099018, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 217400.0'),
 Text(0.631573202190062, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nva
lue = 151700.0'),
 Text(0.6345147335099018, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 345600.0'),
 Text(0.6694454179329997, 0.8035714285714286, 'crime_rate <= 15.5\nsquared_error =
12610818698.225\nsamples = 13\nvalue = 355576.923'),
 Text(0.6602531325585003, 0.7678571428571429, 'employment_rate <= 98.35\nsquared_e
rror = 10296299669.422\nsamples = 11\nvalue = 380531.818'),
 Text(0.6506931557690209, 0.7321428571428571, 'population_density <= 8.5\nsquared_
error = 6071006224.49\nsamples = 7\nvalue = 439214.286'),
 Text(0.6433393274694212, 0.6964285714285714, 'employment_rate <= 96.5\nsquared_er
ror = 3699593888.889\nsamples = 3\nvalue = 374116.667'),
 Text(0.6403977961495815, 0.6607142857142857, 'employment_rate <= 93.15\nsquared_e
rror = 600250000.0\nsamples = 2\nvalue = 333500.0'),
 Text(0.6374562648297416, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 35800
0.0'),
 Text(0.6433393274694212, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 30900
0.0'),
 Text(0.6462808587892611, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 455350.0'),
 Text(0.6580469840686204, 0.6964285714285714, 'median_household_income <= 62869.29
9\nsquared_error = 2287584218.75\nsamples = 4\nvalue = 488037.5'),
 Text(0.6521639214289408, 0.6607142857142857, 'crime_rate <= 5.5\nsquared_error =
128255625.0\nsamples = 2\nvalue = 455675.0'),
 Text(0.649222390109101, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 44435
0.0'),
 Text(0.6551054527487806, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 46700
0.0'),
 Text(0.6639300467083, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 10975
16.188\nsquared_error = 2352250000.0\nsamples = 2\nvalue = 520400.0'),
 Text(0.6609885153884603, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 47190
0.0'),
 Text(0.6668715780281399, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 56890
0.0'),
 Text(0.6698131093479797, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 93
5916.812\nsquared_error = 1118084218.75\nsamples = 4\nvalue = 277837.5'),
 Text(0.6668715780281399, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
```

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alue = 335000.0'),
 Text(0.6727546406678195, 0.6964285714285714, 'median_household_income <= 62725.9
\nsquared_error = 38533888.889\nsamples = 3\nvalue = 258783.333'),
 Text(0.6698131093479797, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 267550.0'),
 Text(0.6756961719876594, 0.6607142857142857, 'population_density <= 8.5\nsquared_
error = 160000.0\nsamples = 2\nvalue = 254400.0'),
 Text(0.6727546406678195, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 25480
0.0'),
 Text(0.6786377033074992, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 25400
0.0'),
 Text(0.6786377033074992, 0.7678571428571429, 'crime_rate <= 18.5\nsquared_error =
3077475625.0\nsamples = 2\nvalue = 218325.0'),
 Text(0.6756961719876594, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 162850.0'),
 Text(0.681579234627339, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nva
lue = 273800.0'),
 Text(0.8188085649118977, 0.9107142857142857, 'jobs_accessable_by_45_minutes <= 90
5274.094\nsquared_error = 36447276804.717\nsamples = 98\nvalue = 398452.551'),
 Text(0.7238637473500364, 0.875, 'population_density <= 2.5\nsquared_error = 16175
595955.556\nsamples = 30\nvalue = 246606.667'),
 Text(0.6977576568864581, 0.8392857142857143, 'population_density <= 0.5\nsquared_
error = 1283899843.75\nsamples = 8\nvalue = 128962.5'),
 Text(0.6948161255666182, 0.8035714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 208800.0'),
 Text(0.7006991882062978, 0.8035714285714286, 'median_household_income <= 114951.8
01\nsquared_error = 426656734.694\nsamples = 7\nvalue = 117557.143'),
 Text(0.6933453599066983, 0.7678571428571429, 'median_household_income <= 104926.0
\nsquared_error = 143177600.0\nsamples = 5\nvalue = 106580.0'),
 Text(0.6874622972670187, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 66
4566.5\nsquared_error = 81686666.667\nsamples = 3\nvalue = 98700.0'),
 Text(0.6845207659471788, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 86000.0'),
 Text(0.6904038285868584, 0.6964285714285714, 'employment_rate <= 95.8\nsquared_er
ror = 1562500.0\nsamples = 2\nvalue = 105050.0'),
 Text(0.6874622972670187, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 103800.0'),
 Text(0.6933453599066983, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 106300.0'),
 Text(0.699228422546378, 0.7321428571428571, 'median_household_income <= 107739.10
2\nsquared_error = 2560000.0\nsamples = 2\nvalue = 118400.0'),
 Text(0.6962868912265381, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 120000.0'),
 Text(0.7021699538662177, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 116800.0'),
 Text(0.7080530165058975, 0.7678571428571429, 'crime_rate <= 0.5\nsquared_error =
81000000.0\nsamples = 2\nvalue = 145000.0'),
 Text(0.7051114851860576, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 136000.0'),
 Text(0.7109945478257372, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 154000.0'),
 Text(0.7499698378136149, 0.8392857142857143, 'crime_rate <= 0.5\nsquared_error =
14727873904.959\nsamples = 22\nvalue = 289386.364'),
 Text(0.731585267064616, 0.8035714285714286, 'median_household_income <= 105753.10
2\nsquared_error = 12385795591.716\nsamples = 13\nvalue = 223719.231'),
 Text(0.7227606731050965, 0.7678571428571429, 'employment_rate <= 96.9\nsquared_er
```

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ror = 10646711734.694\nsamples = 7\nvalue = 286157.143'),
 Text(0.7168776104654169, 0.7321428571428571, 'employment_rate <= 96.3\nsquared_er
ror = 3968057968.75\nsamples = 4\nvalue = 338612.5'),
 Text(0.7139360791455771, 0.6964285714285714, 'population_density <= 5.0\nsquared_
error = 3054242222.222\nsamples = 3\nvalue = 314966.667'),
 Text(0.7109945478257372, 0.6607142857142857, 'employment_rate <= 93.45\nsquared_e
rror = 951722500.0\nsamples = 2\nvalue = 349750.0'),
 Text(0.7080530165058975, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 38060
0.0'),
 Text(0.7139360791455771, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 31890
0.0'),
 Text(0.7168776104654169, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 245400.0'),
 Text(0.7198191417852567, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 409550.0'),
 Text(0.7286437357447761, 0.7321428571428571, 'employment_rate <= 99.45\nsquared_e
rror = 10991160555.556\nsamples = 3\nvalue = 216216.667'),
 Text(0.7257022044249364, 0.6964285714285714, 'population_density <= 4.5\nsquared_
error = 841000000.0\nsamples = 2\nvalue = 144000.0'),
 Text(0.7227606731050965, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 173000.0'),
 Text(0.7286437357447761, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 115000.0'),
 Text(0.731585267064616, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nva
lue = 360650.0'),
 Text(0.7404098610241355, 0.7678571428571429, 'jobs_accessable_by_45_minutes <= 41
5750.0\nsquared_error = 4560203125.0\nsamples = 6\nvalue = 150875.0'),
 Text(0.7374683297042957, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 270000.0'),
 Text(0.7433513923439753, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 77
6649.0\nsquared_error = 2066460000.0\nsamples = 5\nvalue = 127050.0'),
 Text(0.7374683297042957, 0.6964285714285714, 'median_household_income <= 111929.1
99\nsquared_error = 1348763888.889\nsamples = 3\nvalue = 155083.333'),
 Text(0.7345267983844559, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 104000.0'),
 Text(0.7404098610241355, 0.6607142857142857, 'employment_rate <= 94.65\nsquared_e
rror = 66015625.0\nsamples = 2\nvalue = 180625.0'),
 Text(0.7374683297042957, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 17250
0.0'),
 Text(0.7433513923439753, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 18875
0.0'),
 Text(0.7492344549836549, 0.6964285714285714, 'median_household_income <= 113118.5
\nsquared_error = 196000000.0\nsamples = 2\nvalue = 85000.0'),
 Text(0.7462929236638152, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 99000.0'),
 Text(0.7521759863034948, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 71000.0'),
 Text(0.7683544085626138, 0.8035714285714286, 'jobs_accessable_by_45_minutes <= 75
0738.594\nsquared_error = 2885180987.654\nsamples = 9\nvalue = 384238.889'),
 Text(0.7610005802630143, 0.7678571428571429, 'crime_rate <= 2.5\nsquared_error =
206055000.0\nsamples = 3\nvalue = 437950.0'),
 Text(0.7580590489431744, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 71
9283.688\nsquared_error = 62015625.0\nsamples = 2\nvalue = 447025.0'),
 Text(0.7551175176233346, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 439150.0'),
 Text(0.7610005802630143, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
```

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alue = 454900.0'),
 Text(0.7639421115828541, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 419800.0'),
 Text(0.7757082368622134, 0.7678571428571429, 'jobs_accessable_by_45_minutes <= 84
1129.219\nsquared_error = 2061081388.889\nsamples = 6\nvalue = 357383.333'),
 Text(0.7698251742225337, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 78
0321.719\nsquared_error = 360496875.0\nsamples = 4\nvalue = 328225.0'),
 Text(0.766883642902694, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nva
lue = 360700.0'),
 Text(0.7727667055423736, 0.6964285714285714, 'crime_rate <= 2.5\nsquared_error =
11940000.0\nsamples = 3\nvalue = 317400.0'),
 Text(0.7698251742225337, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 312900.0'),
 Text(0.7757082368622134, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 80
5582.719\nsquared_error = 2722500.0\nsamples = 2\nvalue = 319650.0'),
 Text(0.7727667055423736, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 32130
0.0'),
 Text(0.7786497681820532, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 31800
0.0'),
 Text(0.781591299501893, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 869
577.406\nsquared_error = 361000000.0\nsamples = 2\nvalue = 415700.0'),
 Text(0.7786497681820532, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 434700.0'),
 Text(0.7845328308217329, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 396700.0'),
 Text(0.9137533824737589, 0.875, 'median_household_income <= 115839.297\nsquared_e
rror = 30730611904.736\nsamples = 68\nvalue = 465443.382'),
 Text(0.874203574649975, 0.8392857142857143, 'employment_rate <= 99.4\nsquared_err
or = 26085938949.0\nsamples = 50\nvalue = 423001.0'),
 Text(0.8292992605955452, 0.8035714285714286, 'population_density <= 2.0\nsquared_
error = 15164453112.245\nsamples = 42\nvalue = 391371.429'),
 Text(0.799240487420932, 0.7678571428571429, 'median_household_income <= 95461.0\n
squared_error = 9474714400.0\nsamples = 5\nvalue = 233240.0'),
 Text(0.7933574247812524, 0.7321428571428571, 'median_household_income <= 87052.39
8\nsquared_error = 1465215555.556\nsamples = 3\nvalue = 160233.333'),
 Text(0.7904158934614125, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 204200.0'),
 Text(0.7962989561010921, 0.6964285714285714, 'employment_rate <= 98.15\nsquared_e
rror = 748022500.0\nsamples = 2\nvalue = 138250.0'),
 Text(0.7933574247812524, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 110900.0'),
 Text(0.799240487420932, 0.6607142857142857, 'squared_error = 0.0\nsamples = 1\nva
lue = 165600.0'),
 Text(0.8051235500606116, 0.7321428571428571, 'employment_rate <= 98.2\nsquared_er
ror = 1501562500.0\nsamples = 2\nvalue = 342750.0'),
 Text(0.8021820187407718, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 304000.0'),
 Text(0.8080650813804514, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 381500.0'),
 Text(0.8593580337701584, 0.7678571428571429, 'median_household_income <= 84263.10
2\nsquared_error = 12097569572.681\nsamples = 37\nvalue = 412740.541'),
 Text(0.8564165024503185, 0.7321428571428571, 'squared_error = 0.0\nsamples = 1\nv
alue = 691500.0'),
 Text(0.8622995650899982, 0.7321428571428571, 'jobs_accessable_by_45_minutes <= 11
12209.938\nsquared_error = 10215130964.506\nsamples = 36\nvalue = 404997.222'),
 Text(0.8297588748642701, 0.6964285714285714, 'jobs_accessable_by_45_minutes <= 10
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57801.5\nsquared_error = 11077813815.104\nsamples = 24\nvalue = 430943.75'),
 Text(0.8051235500606116, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 92
0727.094\nsquared_error = 5994637291.667\nsamples = 18\nvalue = 387225.0'),
 Text(0.8021820187407718, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 54680
0.0'),
 Text(0.8080650813804514, 0.625, 'jobs_accessable_by_45_minutes <= 967994.0\nsquar
ed_error = 4761258685.121\nsamples = 17\nvalue = 377838.235'),
 Text(0.7926220419512924, 0.5892857142857143, 'population_density <= 4.5\nsquared_
error = 3581444506.173\nsamples = 9\nvalue = 345977.778'),
 Text(0.7837974479917729, 0.5535714285714286, 'jobs_accessable_by_45_minutes <= 93
9191.094\nsquared_error = 3976820468.75\nsamples = 4\nvalue = 305587.5'),
 Text(0.7808559166719331, 0.5178571428571429, 'jobs_accessable_by_45_minutes <= 92
5960.781\nsquared_error = 1250043888.889\nsamples = 3\nvalue = 337416.667'),
 Text(0.7779143853520932, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 385800.0'),
 Text(0.7837974479917729, 0.48214285714285715, 'median_household_income <= 90631.1
02\nsquared_error = 119355625.0\nsamples = 2\nvalue = 313225.0'),
 Text(0.7808559166719331, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 302300.0'),
 Text(0.7867389793116127, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 324150.0'),
 Text(0.7867389793116127, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 210100.0'),
 Text(0.8014466359108119, 0.5535714285714286, 'population_density <= 8.5\nsquared_
error = 915964400.0\nsamples = 5\nvalue = 378290.0'),
 Text(0.798505104590972, 0.5178571428571429, 'employment_rate <= 98.9\nsquared_err
or = 158294218.75\nsamples = 4\nvalue = 392337.5'),
 Text(0.7955635732711323, 0.48214285714285715, 'crime_rate <= 1.5\nsquared_error =
27682222.222\nsamples = 3\nvalue = 385566.667'),
 Text(0.7926220419512924, 0.44642857142857145, 'crime_rate <= 0.5\nsquared_error =
6502500.0\nsamples = 2\nvalue = 382150.0'),
 Text(0.7896805106314525, 0.4107142857142857, 'squared_error = 0.0\nsamples = 1\nv
alue = 379600.0'),
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alue = 384700.0'),
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alue = 392400.0'),
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alue = 322100.0'),
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33368.5\nsquared_error = 3661853085.938\nsamples = 8\nvalue = 413681.25'),
 Text(0.8161542925100109, 0.5535714285714286, 'median_household_income <= 103247.9
02\nsquared_error = 2802497847.222\nsamples = 6\nvalue = 436241.667'),
 Text(0.8102712298703313, 0.5178571428571429, 'employment_rate <= 97.65\nsquared_e
rror = 877801666.667\nsamples = 3\nvalue = 477450.0'),
 Text(0.8073296985504915, 0.48214285714285715, 'median_household_income <= 93010.8
98\nsquared_error = 301022500.0\nsamples = 2\nvalue = 459050.0'),
 Text(0.8043881672306517, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 476400.0'),
 Text(0.8102712298703313, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 441700.0'),
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value = 514250.0'),
 Text(0.8220373551496907, 0.5178571428571429, 'median_household_income <= 104730.8
```

```
01\nsquared_error = 1330940555.556\nsamples = 3\nvalue = 395033.333'),
 Text(0.8190958238298508, 0.48214285714285715, 'squared_error = 0.0\nsamples = 1\n
value = 346000.0'),
 Text(0.8249788864695304, 0.48214285714285715, 'population_density <= 3.5\nsquared
_error = 193210000.0\nsamples = 2\nvalue = 419550.0'),
 Text(0.8220373551496907, 0.44642857142857145, 'squared_error = 0.0\nsamples = 1\n
value = 405650.0'),
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value = 433450.0'),
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 Text(0.8279204177893703, 0.5178571428571429, 'squared_error = 0.0\nsamples = 1\nv
alue = 334500.0'),
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ue = 357500.0'),
 Text(0.8543941996679287, 0.6607142857142857, 'median_household_income <= 101656.1
02\nsquared_error = 3391394166.667\nsamples = 6\nvalue = 562100.0'),
 Text(0.8485111370282491, 0.625, 'jobs_accessable_by_45_minutes <= 1095796.812\nsq
uared_error = 1073769218.75\nsamples = 4\nvalue = 540237.5'),
 Text(0.8455696057084092, 0.5892857142857143, 'employment_rate <= 97.5\nsquared_er
ror = 468107222.222\nsamples = 3\nvalue = 524716.667'),
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\nsquared_error = 150675625.0\nsamples = 2\nvalue = 538275.0'),
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alue = 526000.0'),
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alue = 550550.0'),
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alue = 497600.0'),
 Text(0.8514526683480889, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 586800.0'),
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rror = 5158830625.0\nsamples = 2\nvalue = 605825.0'),
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alue = 677650.0'),
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alue = 534000.0'),
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 Text(0.8793972158865672, 0.6607142857142857, 'jobs_accessable_by_45_minutes <= 11
56368.25\nsquared_error = 407380585.938\nsamples = 8\nvalue = 316581.25'),
 Text(0.8720433875869676, 0.625, 'population_density <= 4.0\nsquared_error = 13225
00.0\nsamples = 2\nvalue = 292050.0'),
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alue = 293200.0'),
 Text(0.8749849189068074, 0.5892857142857143, 'squared_error = 0.0\nsamples = 1\nv
alue = 290900.0'),
 Text(0.8867510441861668, 0.625, 'median_household_income <= 95765.898\nsquared_er
ror = 275274513.889\nsamples = 6\nvalue = 324758.333'),
 Text(0.8808679815464872, 0.5892857142857143, 'median_household_income <= 89910.89
8\nsquared_error = 16453888.889\nsamples = 3\nvalue = 309483.333'),
 Text(0.8779264502266473, 0.5535714285714286, 'squared_error = 0.0\nsamples = 1\nv
alue = 315050.0'),
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8\nsquared_error = 1440000.0\nsamples = 2\nvalue = 306700.0'),
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```

```
alue = 305500.0'),
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alue = 329750.0'),
 Text(0.8955756381456862, 0.5535714285714286, 'employment_rate <= 97.4\nsquared_er
ror = 21855625.0\nsamples = 2\nvalue = 345175.0'),
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alue = 340500.0'),
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alue = 349850.0'),
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3\nsquared_error = 4532992500.0\nsamples = 4\nvalue = 426150.0'),
 Text(0.9073417634250456, 0.625, 'jobs_accessable_by_45_minutes <= 1255688.188\nsq
uared_error = 17055555.556\nsamples = 3\nvalue = 387333.333'),
 Text(0.9044002321052057, 0.5892857142857143, 'crime_rate <= 99.5\nsquared_error =
62500.0\nsamples = 2\nvalue = 390250.0'),
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0.0'),
 Text(0.9191078887044049, 0.8035714285714286, 'employment_rate <= 99.75\nsquared_e
rror = 50597137148.438\nsamples = 8\nvalue = 589056.25'),
 Text(0.916166357384565, 0.7678571428571429, 'squared_error = 0.0\nsamples = 1\nva
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2\nsquared_error = 20452496020.408\nsamples = 7\nvalue = 520707.143'),
 Text(0.916166357384565, 0.7321428571428571, 'population_density <= 6.5\nsquared_e
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alue = 742000.0'),
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25.0\nsamples = 2\nvalue = 501175.0'),
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alue = 375300.0'),
 Text(0.9308740139837641, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
```

```
alue = 291200.0'),
 Text(0.9533031902975427, 0.8392857142857143, 'median_household_income <= 117016.1
99\nsquared_error = 24729364598.766\nsamples = 18\nvalue = 583338.889'),
 Text(0.950361658977703, 0.8035714285714286, 'squared_error = 0.0\nsamples = 1\nva
lue = 1000200.0'),
 Text(0.9562447216173826, 0.8035714285714286, 'population_density <= 6.5\nsquared_
error = 15360789688.581\nsamples = 17\nvalue = 558817.647'),
 Text(0.9477878190728432, 0.7678571428571429, 'median_household_income <= 121544.5
98\nsquared_error = 9770704222.222\nsamples = 15\nvalue = 530316.667'),
 Text(0.9396986079432836, 0.7321428571428571, 'population_density <= 5.0\nsquared_
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alue = 411700.0'),
 Text(0.9426401392631234, 0.6964285714285714, 'squared_error = 0.0\nsamples = 1\nv
alue = 340550.0'),
 Text(0.9558770302024027, 0.7321428571428571, 'employment_rate <= 98.5\nsquared_er
ror = 6858758136.095\nsamples = 13\nvalue = 554038.462'),
 Text(0.948523201902803, 0.6964285714285714, 'median_household_income <= 144869.29
7\nsquared_error = 5260759669.421\nsamples = 11\nvalue = 572881.818'),
 Text(0.9426401392631234, 0.6607142857142857, 'employment_rate <= 90.15\nsquared_e
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941630468.75\nsamples = 4\nvalue = 546662.5'),
 Text(0.9308740139837641, 0.5178571428571429, 'crime_rate <= 0.5\nsquared_error =
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alue = 545750.0'),
 Text(0.9544062645424827, 0.6607142857142857, 'employment_rate <= 95.7\nsquared_er
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 Text(0.9514647332226429, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 57410
```

```
0.0'),
 Text(0.9573477958623225, 0.625, 'squared_error = 0.0\nsamples = 1\nvalue = 73850
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alue = 846750.0'),
 Text(0.9867631090607208, 0.9464285714285714, 'median_household_income <= 184883.1
02\nsquared_error = 217887828580.247\nsamples = 9\nvalue = 1045205.556'),
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 Text(0.9794092807611212, 0.9107142857142857, 'crime_rate <= 1.5\nsquared_error =
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alue = 837800.0'),
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error = 199509570555.556\nsamples = 3\nvalue = 1582483.333'),
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2500.0\nsamples = 2\nvalue = 1895150.0'),
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alue = 1817800.0'),
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```

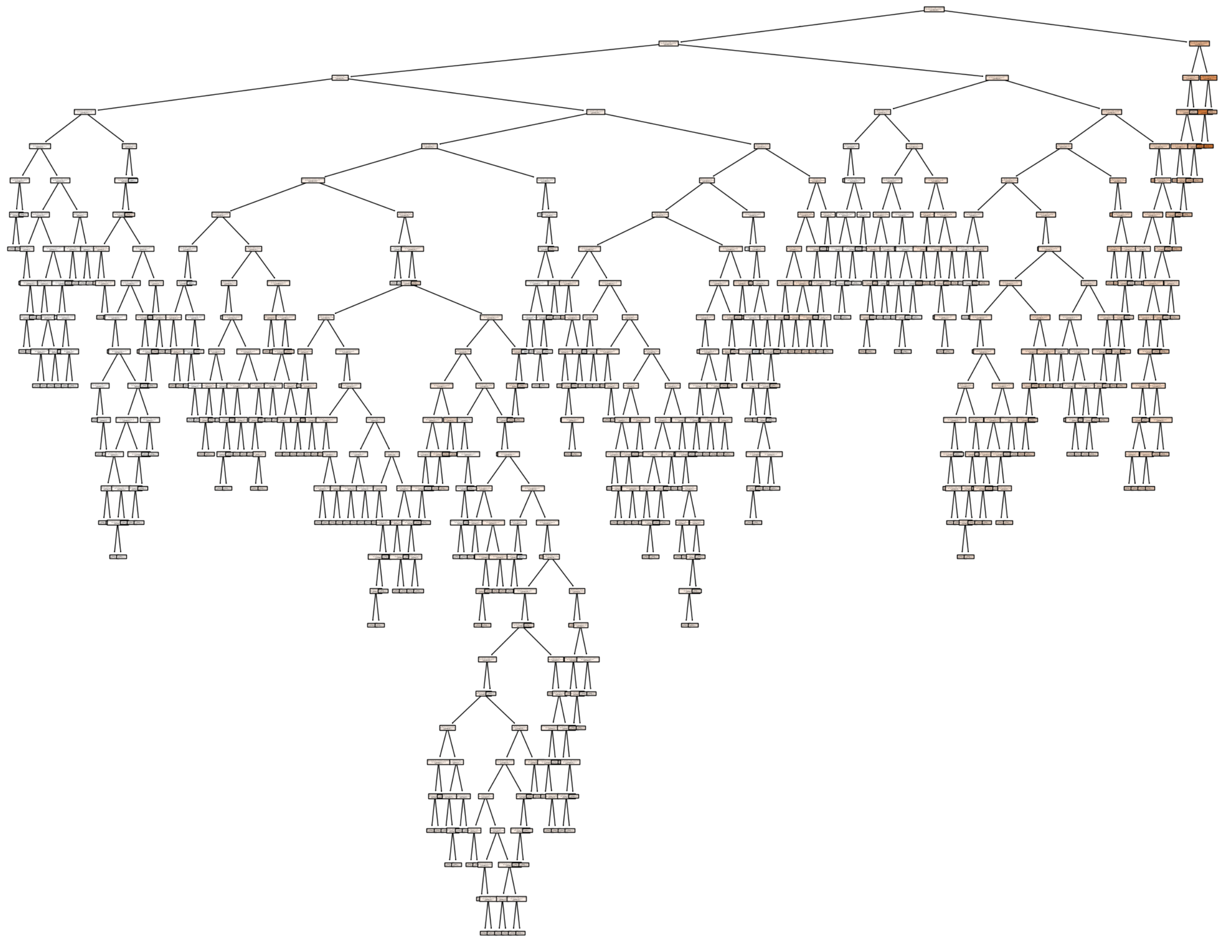

In [30]:
```
# initiate a decision tree with a max depth of 4
dt_pre_pruned = DecisionTreeRegressor(max_depth=5, min_samples_split=6, min_samples_

# Train the model
dt_pre_pruned.fit(X_train, y_train)

rmse = np.sqrt(mean_squared_error(y_test, predictions))
print(f"Mean Squared Error: {rmse}")

fig = plt.figure(figsize=(25,20))
tree.plot_tree(dt_pre_pruned,
                   feature_names=['jobs_accessable_by_45_minutes', 'median_househol
                   filled=True)
```

```
Mean Squared Error: 168018.25995810048
```

```
Out[30]: [Text(0.7058823529411765, 0.9166666666666666, 'median_household_income <= 160051.8
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 Text(0.2647058823529412, 0.08333333333333333, 'squared_error = 8144606557.007\nsa
mples = 128\nvalue = 227927.344'),
 Text(0.3235294117647059, 0.08333333333333333, 'squared_error = 6606907222.222\nsa
mples = 9\nvalue = 124666.667'),
 Text(0.4117647058823529, 0.25, 'population_density <= 7.5\nsquared_error = 801113
2197.022\nsamples = 76\nvalue = 277822.368'),
 Text(0.38235294117647056, 0.08333333333333333, 'squared_error = 5557021649.03\nsa
mples = 63\nvalue = 261777.778'),
 Text(0.4411764705882353, 0.08333333333333333, 'squared_error = 12610818698.225\ns
amples = 13\nvalue = 355576.923'),
 Text(0.7058823529411765, 0.5833333333333334, 'jobs_accessable_by_45_minutes <= 90
5274.094\nsquared_error = 36447276804.717\nsamples = 98\nvalue = 398452.551'),
 Text(0.5882352941176471, 0.4166666666666667, 'population_density <= 2.5\nsquared_
error = 16175595955.556\nsamples = 30\nvalue = 246606.667'),
 Text(0.5294117647058824, 0.25, 'employment_rate <= 98.15\nsquared_error = 1283899
843.75\nsamples = 8\nvalue = 128962.5'),
 Text(0.5, 0.08333333333333333, 'squared_error = 496605833.333\nsamples = 6\nvalue
= 117150.0'),
 Text(0.5588235294117647, 0.08333333333333333, 'squared_error = 1971360000.0\nsamp
les = 2\nvalue = 164400.0'),
 Text(0.6470588235294118, 0.25, 'crime_rate <= 0.5\nsquared_error = 14727873904.95
9\nsamples = 22\nvalue = 289386.364'),
 Text(0.6176470588235294, 0.08333333333333333, 'squared_error = 12385795591.716\ns
amples = 13\nvalue = 223719.231'),
 Text(0.6764705882352942, 0.08333333333333333, 'squared_error = 2885180987.654\nsa
mples = 9\nvalue = 384238.889'),
 Text(0.8235294117647058, 0.4166666666666667, 'median_household_income <= 115839.2
97\nsquared_error = 30730611904.736\nsamples = 68\nvalue = 465443.382'),
 Text(0.7647058823529411, 0.25, 'employment_rate <= 99.4\nsquared_error = 26085938
949.0\nsamples = 50\nvalue = 423001.0'),
 Text(0.7352941176470589, 0.08333333333333333, 'squared_error = 15164453112.245\ns
```
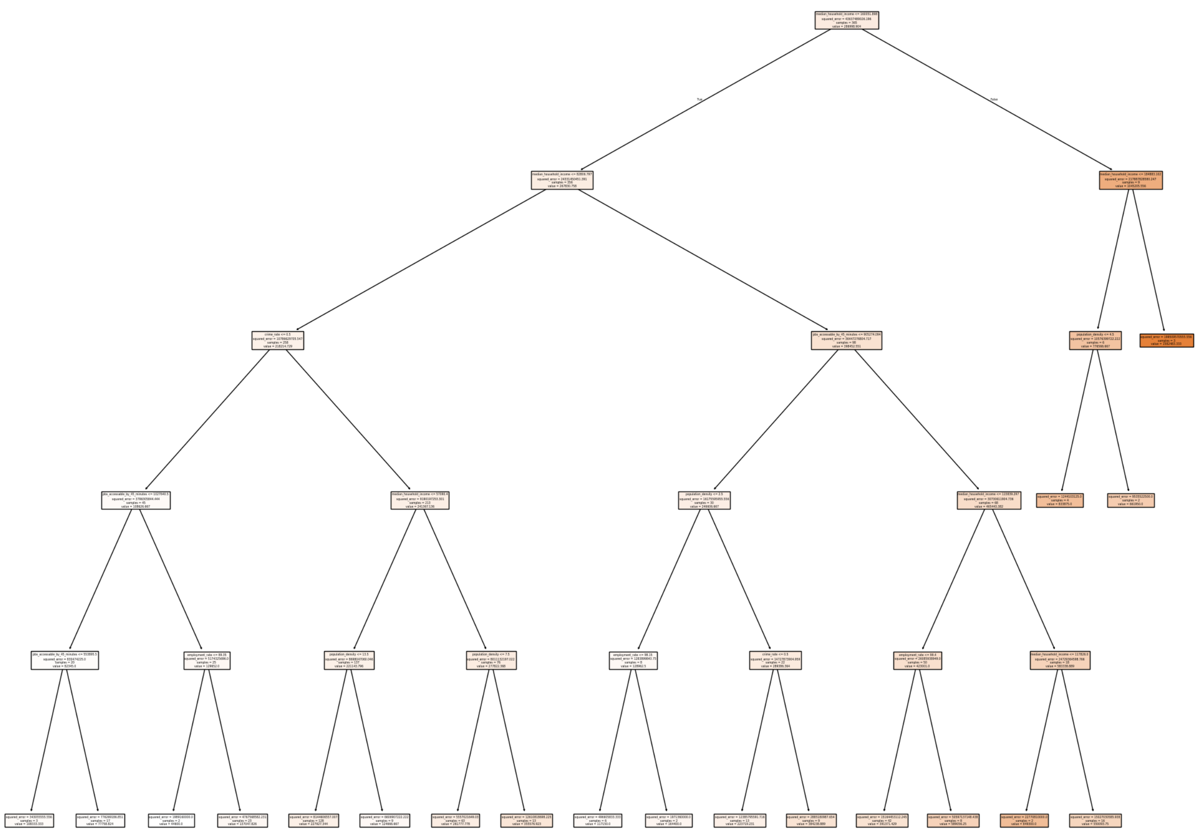

```
amples = 42\nvalue = 391371.429'),
 Text(0.7941176470588235, 0.08333333333333333, 'squared_error = 50597137148.438\ns
amples = 8\nvalue = 589056.25'),
 Text(0.8823529411764706, 0.25, 'median_household_income <= 117826.0\nsquared_erro
r = 24729364598.766\nsamples = 18\nvalue = 583338.889'),
 Text(0.8529411764705882, 0.08333333333333333, 'squared_error = 22770810000.0\nsam
ples = 2\nvalue = 849300.0'),
 Text(0.9117647058823529, 0.08333333333333333, 'squared_error = 15027030585.938\ns
amples = 16\nvalue = 550093.75'),
 Text(0.9411764705882353, 0.75, 'median_household_income <= 184883.102\nsquared_er
ror = 217887828580.247\nsamples = 9\nvalue = 1045205.556'),
 Text(0.8235294117647058, 0.8333333333333333, '  False'),
 Text(0.9117647058823529, 0.5833333333333334, 'population_density <= 4.5\nsquared_
error = 10576399722.222\nsamples = 6\nvalue = 776566.667'),
 Text(0.8823529411764706, 0.4166666666666667, 'squared_error = 1244103125.0\nsampl
es = 4\nvalue = 833875.0'),
 Text(0.9411764705882353, 0.4166666666666667, 'squared_error = 9535522500.0\nsampl
es = 2\nvalue = 661950.0'),
 Text(0.9705882352941176, 0.5833333333333334, 'squared_error = 199509570555.556\ns
amples = 3\nvalue = 1582483.333')]
```



In [31]:
```
from pygam import (s as s_gam,
                   l as l_gam,
                   f as f_gam,
                   LinearGAM)

gam_full = LinearGAM(s_gam(0) +
```

```
                     s_gam(1, n_splines=7) +
                     f_gam(2, lam=0))
Xgam = np.column_stack([X])
gam_full = gam_full.fit(Xgam, y)
```

In [32]:
```
gam_full.summary()
```

```
LinearGAM
=============================================== ==========================================================
Distribution:                        NormalDist Effective DoF:
37.2594
Link Function:                     IdentityLink Log Likelihood:
-6038.4306
Number of Samples:                          457 AIC:
12153.38
                                                AICc:
12160.5712
                                                GCV:
22289854140.2677
                                                Scale:
138001.8931
                                                Pseudo R-Squared:
0.6063
==========================================================================================================
Feature Function                  Lambda               Rank         EDoF         P >
x         Sig. Code
================================= ==================== ============ ============ ===
========= ============
s(0)                              [0.6]                20           11.5         1.9
8e-07     ***
s(1)                              [0.6]                7            3.1          1.1
1e-16     ***
f(2)                              [0]                  57           22.6         8.4
0e-12     ***
intercept                                              1            0.0          1.1
1e-16     ***
==========================================================================================================
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

WARNING: Fitting splines and a linear function to a feature introduces a model ident
ifiability problem
         which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models
or models with
         known smoothing parameters, but when smoothing parameters have been estimat
ed, the p-values
         are typically lower than they should be, meaning that the tests reject the
null too readily.
```

```
C:\Users\Isaac\AppData\Local\Temp\ipykernel_4692\3667082475.py:1: UserWarning: KNOWN
BUG: p-values computed in this summary are likely much smaller than they should be.

Please do not make inferences based on these values!

Collaborate on a solution, and stay up to date at:
github.com/dswah/pyGAM/issues/163

  gam_full.summary()
```